

Classical Theory of Linear Models

Slide edition

A complete treatment of OLS, Gauss–Markov,
and exact inference under normality

1 The Multiple Linear Regression Model

1.1 Model Specification and Matrix Notation

Definition 1.1 (Multiple Linear Regression Model). Given n observations, each consisting of a dependent variable Y_i and k independent variables $X_{i1}, \dots, X_{ik}$, the **multiple linear regression model** assumes

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n, \quad (1)$$

where β_0 is the intercept, $\beta_1, \dots, \beta_k$ are slope coefficients, and ε_i is the error term. Equivalently, in matrix form,

$$\boxed{\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}} \quad (2)$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}_{n \times 1}, \quad \mathbf{X} = \begin{pmatrix} 1 & X_{11} & \cdots & X_{1k} \\ 1 & X_{21} & \cdots & X_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{nk} \end{pmatrix}_{n \times (k+1)}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}_{(k+1) \times 1}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}_{n \times 1}.$$

The matrix $\mathbf{X}$ is called the **design matrix**. Its first column is the all-ones vector $\mathbf{1} = (1, 1, \dots, 1)'$, corresponding to the intercept. We write $\mathbf{x}'_i$ for the i -th row of $\mathbf{X}$, so that $Y_i = \mathbf{x}'_i \boldsymbol{\beta} + \varepsilon_i$.

1.2 Classical Assumptions

Definition 1.2 (Classical Assumptions). The following assumptions are imposed on model (2):

(A1) **Linearity:** $E[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$.

(A2) **Zero conditional mean:** $E[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$.

(A3) **Spherical errors:** $\text{var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$ for a constant $\sigma^2 > 0$. This comprises two sub-conditions:

- *Homoscedasticity:* $\text{var}(\varepsilon_i) = \sigma^2$ for all i (a constant).
- *No serial correlation:* $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$.

(A4) **Full rank:** $\text{rank}(\mathbf{X}) = k + 1$ (the columns of $\mathbf{X}$ are linearly independent). This requires $n \geq k + 1$. For variance estimation and inference (Sections 3–7 and 9), we additionally require $n > k + 1$ so that $\text{MSE} = \text{SSE}/(n - k - 1)$ (see Definition 3.1) is well-defined.

(A5) **Normality:** $\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

Assumptions (A1)–(A4) are called the **Gauss–Markov assumptions**. Assumption (A5) is additionally needed for exact distributional results in finite samples.

Remark. Throughout this article, we treat $\mathbf{X}$ as fixed (non-stochastic) and condition on it. All expectations, variances, and distributions are conditional on $\mathbf{X}$, but we suppress the conditioning notation for brevity.

Remark (Relationship between (A1) and (A2)). Given the model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, assumptions (A1) and (A2) are logically equivalent: taking conditional expectations of both sides gives $E[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta} + E[\boldsymbol{\varepsilon} \mid \mathbf{X}]$, so $E[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$ holds if and only if $E[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$. We state both because each form is convenient in different

proofs: (A1) is closer to how the model is written, and (A2) is closer to how residuals are handled. Conditioning on $\mathbf{X}$ (previous remark) also means that inside any proof, $E[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ can be used without further notation—these are the conditional expectation and variance with the conditioning suppressed.

2 Ordinary Least Squares

2.1 Definition and the Normal Equations

Definition 2.1 (Ordinary Least Squares). The **OLS estimator** $\hat{\boldsymbol{\beta}}$ is the value of $\boldsymbol{\beta}$ that minimizes the sum of squared residuals:

$$Q(\boldsymbol{\beta}) = \sum_{i=1}^n (Y_i - \mathbf{x}'_i \boldsymbol{\beta})^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2. \quad (3)$$

Lemma 2.1 (Matrix Differentiation Formulas). For $\boldsymbol{\beta} \in \mathbb{R}^p$, constant $\mathbf{a} \in \mathbb{R}^p$ and constant $\mathbf{A} \in \mathbb{R}^{p \times p}$, write $Df(\boldsymbol{\beta}) := \partial f / \partial \boldsymbol{\beta}'$ (numerator layout, row vector). Then

- (i) $D(\mathbf{a}'\boldsymbol{\beta}) = D(\boldsymbol{\beta}'\mathbf{a}) = \mathbf{a}'$;
- (ii) $D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')$, which equals $2\boldsymbol{\beta}'\mathbf{A}$ when $\mathbf{A}' = \mathbf{A}$.

Proof. Apply the first identification theorem: $df(\boldsymbol{\beta}) = \mathbf{c}(\boldsymbol{\beta})'d\boldsymbol{\beta} \iff Df(\boldsymbol{\beta}) = \mathbf{c}(\boldsymbol{\beta})'$.

(i). $d(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'d\boldsymbol{\beta}$ (since $d\mathbf{a} = 0$), so $D(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'$; the case $D(\boldsymbol{\beta}'\mathbf{a})$ follows from $\boldsymbol{\beta}'\mathbf{a} = \mathbf{a}'\boldsymbol{\beta}$.

(ii). Since $d\mathbf{A} = 0$, the product rule and the scalar identity $(d\boldsymbol{\beta})'\mathbf{A}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{A}'d\boldsymbol{\beta}$ give

$$d(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = (d\boldsymbol{\beta})'\mathbf{A}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{A}d\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{A}'d\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{A}d\boldsymbol{\beta} = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')d\boldsymbol{\beta}.$$

□

2.2 Derivation via Differentiation

Theorem 2.2 (OLS Estimator). *Under assumption (A4), the OLS estimator is*

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.} \quad (4)$$

Proof. **Step 1: Expand $Q(\boldsymbol{\beta})$.**

$$\begin{aligned} Q(\boldsymbol{\beta}) &= (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{Y}'\mathbf{Y} - \mathbf{Y}'\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}'\mathbf{X}'\mathbf{Y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \\ &= \mathbf{Y}'\mathbf{Y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{Y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}. \end{aligned} \quad (5)$$

In the last line, $\mathbf{Y}'\mathbf{X}\boldsymbol{\beta}$ is a scalar, hence equals its transpose $\boldsymbol{\beta}'\mathbf{X}'\mathbf{Y}$.

Step 2: First-order condition. By Lemma 2.1 with $\mathbf{a} = \mathbf{X}'\mathbf{Y}$ and $\mathbf{A} = \mathbf{X}'\mathbf{X}$ (symmetric),

$$\frac{\partial Q}{\partial \boldsymbol{\beta}'} = -2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}.$$

Setting this to $\mathbf{0}'$ and transposing gives the **normal equation**

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}. \quad (6)$$

Step 3: Solve. By (A4), $\text{rank}(\mathbf{X}) = k + 1$. We show $\text{rank}(\mathbf{X}'\mathbf{X}) = \text{rank}(\mathbf{X})$: $\mathbf{X}\mathbf{v} = \mathbf{0}$ trivially implies $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$; conversely, $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 = 0$, hence $\mathbf{X}\mathbf{v} = \mathbf{0}$. So $\mathbf{X}$ and $\mathbf{X}'\mathbf{X}$ have the same null space, hence the same rank. Thus $\mathbf{X}'\mathbf{X}$ is $(k + 1) \times (k + 1)$ of full rank, hence invertible. Multiplying (6) by $(\mathbf{X}'\mathbf{X})^{-1}$:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

Step 4: Verify the second-order condition. The Hessian matrix is

$$\frac{\partial^2 Q}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} = 2\mathbf{X}'\mathbf{X}.$$

Under assumption (A4), for any nonzero $\mathbf{v} \in \mathbb{R}^{k+1}$, $\mathbf{X}\mathbf{v} \neq \mathbf{0}$, so $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 > 0$. Hence $\mathbf{X}'\mathbf{X}$ is positive definite, confirming that $\hat{\boldsymbol{\beta}}$ is a global minimum. $\square$

2.3 Derivation via Orthogonal Projection

Theorem 2.3 (OLS as Orthogonal Projection). *The fitted vector $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$, that is, $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$, and the OLS estimator is $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$.*

Proof. The OLS problem is to minimize $Q(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|^2$ over $\beta \in \mathbb{R}^{k+1}$, equivalently, to find the point in $\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^{k+1}\}$ closest to $\mathbf{Y}$ in Euclidean norm.

Step 1: Orthogonality characterization of the minimum. We claim $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ minimizes $\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2$ over $\tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ iff $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$.

($\Leftarrow$) Assume the orthogonality condition and let $\tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$. Since $\mathcal{C}(\mathbf{X})$ is a subspace, $\hat{\mathbf{Y}} - \tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$, so $(\mathbf{Y} - \hat{\mathbf{Y}}) \perp (\hat{\mathbf{Y}} - \tilde{\mathbf{Y}})$. Pythagoras applied to $\mathbf{Y} - \tilde{\mathbf{Y}} = (\mathbf{Y} - \hat{\mathbf{Y}}) + (\hat{\mathbf{Y}} - \tilde{\mathbf{Y}})$ gives

$$\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 + \|\hat{\mathbf{Y}} - \tilde{\mathbf{Y}}\|^2 \geq \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2,$$

with equality iff $\tilde{\mathbf{Y}} = \hat{\mathbf{Y}}$. So $\hat{\mathbf{Y}}$ is the unique minimizer.

($\Rightarrow$) Let $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ minimize. Suppose, for contradiction, some $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ has $\mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) \neq 0$; normalize to $\|\mathbf{v}\|^2 = 1$. For $\tilde{\mathbf{Y}} = \hat{\mathbf{Y}} + t\mathbf{v} \in \mathcal{C}(\mathbf{X})$,

$$\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 - 2t \mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) + t^2,$$

minimized at $t^* = \mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) \neq 0$ with value $\|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 - (t^*)^2 < \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2$, contradicting optimality. Hence $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$.

Step 2: Translate the orthogonality condition to a normal equation. The residual is orthogonal to every vector in $\mathcal{C}(\mathbf{X})$ if and only if it is orthogonal to every column of $\mathbf{X}$ (since those columns span $\mathcal{C}(\mathbf{X})$):

$$\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X}) \iff \mathbf{X}'(\mathbf{Y} - \hat{\mathbf{Y}}) = \mathbf{0}.$$

Writing $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ for some $\hat{\boldsymbol{\beta}}$, this becomes $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$, the same normal equation (6) obtained by differentiation.

Step 3: Solve. Under assumption (A4), $\mathbf{X}'\mathbf{X}$ is invertible, so $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$, and the minimizer $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ exists and is unique. $\square$

2.4 Hat Matrix, Residual Maker, and Idempotent Matrices

Definition 2.2 (Hat Matrix and Residual Maker Matrix). The **hat matrix** is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'. \quad (7)$$

The **residual maker matrix** (or **annihilator matrix**) is

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}. \quad (8)$$

Then $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{Y}$, and the **residual vector** $\mathbf{e} := \mathbf{Y} - \hat{\mathbf{Y}}$, with components $e_i = Y_i - \hat{Y}_i$, satisfies $\mathbf{e} = \mathbf{M}\mathbf{Y}$.

Definition 2.3 (Idempotent Matrix). A square matrix $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$.

Lemma 2.4 (Trace Equals Rank for Idempotent Matrices). *If $\mathbf{A}$ is an $n \times n$ idempotent matrix, then $\text{tr}(\mathbf{A}) = \text{rank}(\mathbf{A})$.*

Proof. We show every idempotent matrix is diagonalizable with eigenvalues in $\{0, 1\}$.

Step 1: Eigenvalues are in $\{0, 1\}$. If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $\lambda\mathbf{v} = \mathbf{A}\mathbf{v} = \mathbf{A}^2\mathbf{v} = \lambda^2\mathbf{v}$, forcing $\lambda \in \{0, 1\}$.

Step 2: $\mathbb{R}^n = \mathcal{N}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})$. Any $\mathbf{v} \in \mathbb{R}^n$ decomposes as $\mathbf{v} = (\mathbf{v} - \mathbf{A}\mathbf{v}) + \mathbf{A}\mathbf{v}$, with $\mathbf{A}\mathbf{v} \in \mathcal{C}(\mathbf{A})$ and $\mathbf{v} - \mathbf{A}\mathbf{v} \in \mathcal{N}(\mathbf{A})$ (since $\mathbf{A}(\mathbf{v} - \mathbf{A}\mathbf{v}) = \mathbf{A}\mathbf{v} - \mathbf{A}^2\mathbf{v} = \mathbf{0}$). The sum is direct: if $\mathbf{v} = \mathbf{A}\mathbf{w} \in \mathcal{C}(\mathbf{A}) \cap \mathcal{N}(\mathbf{A})$, then $\mathbf{v} = \mathbf{A}\mathbf{w} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{v} = \mathbf{0}$.

Step 3: Diagonalizability. Any $\mathbf{u} \in \mathcal{C}(\mathbf{A})$ satisfies $\mathbf{A}\mathbf{u} = \mathbf{u}$ ($\mathbf{u} = \mathbf{A}\mathbf{w}$, so $\mathbf{A}\mathbf{u} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{w} = \mathbf{u}$). Combining a basis of $\mathcal{N}(\mathbf{A})$ (eigenvalue 0) and a basis of $\mathcal{C}(\mathbf{A})$ (eigenvalue 1) gives an eigenvector basis of $\mathbb{R}^n$ by Step 2. Thus $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$ with $\mathbf{\Lambda}$ diagonal and entries in $\{0, 1\}$.

Step 4: Conclude. Let $r = \text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$. Then $\mathbf{\Lambda}$ has r ones and $n - r$ zeros, so by the cyclic property of trace,

$$\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}) = \text{tr}(\mathbf{\Lambda}) = r = \text{rank}(\mathbf{A}).$$

□

Theorem 2.5 (Properties of $\mathbf{H}$ and $\mathbf{M}$). *The matrices $\mathbf{H}$ and $\mathbf{M}$ satisfy:*

- (i) $\mathbf{H}$ and $\mathbf{M}$ are both symmetric and idempotent.
- (ii) $\mathbf{H}\mathbf{X} = \mathbf{X}$ and $\mathbf{M}\mathbf{X} = \mathbf{0}$.
- (iii) $\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H} = \mathbf{0}$.
- (iv) $\text{tr}(\mathbf{H}) = k + 1$ and $\text{tr}(\mathbf{M}) = n - k - 1$.

Proof. (i) **Symmetry of $\mathbf{H}$:** Since $\mathbf{X}'\mathbf{X}$ is symmetric, so is $(\mathbf{X}'\mathbf{X})^{-1}$:

$$\mathbf{H}' = [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']' = \mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}]'\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}.$$

Idempotency of $\mathbf{H}$:

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \underbrace{\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}}_{\mathbf{I}_{k+1}} \mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' = \mathbf{H}.$$

Symmetry and idempotency of $\mathbf{M}$: $\mathbf{M}' = (\mathbf{I} - \mathbf{H})' = \mathbf{I} - \mathbf{H}' = \mathbf{I} - \mathbf{H} = \mathbf{M}$. Also, $\mathbf{M}^2 = (\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H} = \mathbf{I} - \mathbf{H} = \mathbf{M}$.

(ii) $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{X}\mathbf{I}_{k+1} = \mathbf{X}$, and $\mathbf{M}\mathbf{X} = (\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{X} - \mathbf{X} = \mathbf{0}$.

(iii) $\mathbf{H}\mathbf{M} = \mathbf{H}(\mathbf{I} - \mathbf{H}) = \mathbf{H} - \mathbf{H}^2 = \mathbf{H} - \mathbf{H} = \mathbf{0}$, and $\mathbf{M}\mathbf{H} = (\mathbf{I} - \mathbf{H})\mathbf{H} = \mathbf{H} - \mathbf{H}^2 = \mathbf{0}$.

(iv) By Lemma 2.4, $\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H})$. We show $\text{rank}(\mathbf{H}) = k + 1$ in two steps.

Step 1: $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$. The inclusion $\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$ is immediate: every column of $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is of the form $\mathbf{X}\mathbf{v}$ for some $\mathbf{v} \in \mathbb{R}^{k+1}$. Conversely, $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{H})$ because $\mathbf{H}\mathbf{X} = \mathbf{X}$ by (ii), so every column of $\mathbf{X}$ equals $\mathbf{H}$ times a column of $\mathbf{X}$, hence lies in $\mathcal{C}(\mathbf{H})$.

Step 2: $\dim \mathcal{C}(\mathbf{X}) = k + 1$ by assumption (A4).

Combining, $\text{rank}(\mathbf{H}) = \dim \mathcal{C}(\mathbf{H}) = \dim \mathcal{C}(\mathbf{X}) = k + 1$, so $\text{tr}(\mathbf{H}) = k + 1$. Finally, $\text{tr}(\mathbf{M}) = \text{tr}(\mathbf{I}) - \text{tr}(\mathbf{H}) = n - (k + 1) = n - k - 1$. $\square$

Remark ($\mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$). Combining (i), (ii), and the identity $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ proved above: $\mathbf{H}$ is symmetric, idempotent, and has column space $\mathcal{C}(\mathbf{X})$. These three properties characterize the *orthogonal projection* onto $\mathcal{C}(\mathbf{X})$ (a matrix $\mathbf{P}$ with $\mathbf{P}\mathbf{v} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{v} - \mathbf{P}\mathbf{v} \perp \mathcal{C}(\mathbf{X})$ for every $\mathbf{v}$): any symmetric idempotent $\mathbf{P}$ with $\mathcal{C}(\mathbf{P}) = \mathcal{C}(\mathbf{X})$ equals $\mathbf{H}$. Geometrically, $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ is the closest point in $\mathcal{C}(\mathbf{X})$ to $\mathbf{Y}$ in Euclidean distance (this is the content of §2, specifically the orthogonal projection derivation). Analogously,

$\mathbf{M} = \mathbf{I} - \mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})^\perp$. More generally, for any subspace $\mathcal{S} = \mathcal{C}(\mathbf{B})$ with $\mathbf{B}$ of full column rank, $\mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'$ is the orthogonal projection onto $\mathcal{S}$ (and for $\mathcal{S} = \{\mathbf{0}\}$ it is the zero matrix).

2.5 Errors Versus Residuals

Remark (Errors versus residuals). The **error** $\varepsilon_i = Y_i - \mathbf{x}_i'\boldsymbol{\beta}$ is the deviation from the true (unknown) regression function; it is *unobservable*. The **residual** $e_i = Y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}} = Y_i - \hat{Y}_i$ is the deviation from the fitted regression function; it is *observable*. In matrix form, $\mathbf{e} = \mathbf{M}\mathbf{Y} = \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{M}\boldsymbol{\varepsilon}$ (since $\mathbf{M}\mathbf{X} = \mathbf{0}$ by Theorem 2.5(ii)). The residual vector is a linear transformation of the error vector.

2.6 Algebraic Properties of OLS Residuals

Theorem 2.6 (Implications of the Normal Equations). *The OLS residuals satisfy:*

- (i) $\mathbf{X}'\mathbf{e} = \mathbf{0}$ (residuals are orthogonal to every column of $\mathbf{X}$).
- (ii) $\sum_{i=1}^n e_i = 0$ (residuals sum to zero), provided the model includes an intercept.
- (iii) $\hat{\mathbf{Y}}'\mathbf{e} = 0$ (residuals are orthogonal to fitted values).

Proof. (i) From the normal equation $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$, rearranging:

$$\mathbf{X}'\mathbf{Y} - \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0} \iff \mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0} \iff \mathbf{X}'\mathbf{e} = \mathbf{0}.$$

(ii) The first column of $\mathbf{X}$ is $\mathbf{1}$. The first component of the vector equation $\mathbf{X}'\mathbf{e} = \mathbf{0}$ is $\mathbf{1}'\mathbf{e} = \sum_{i=1}^n e_i = 0$. This relies on the presence of the intercept column.

(iii) Since $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$:

$$\hat{\mathbf{Y}}'\mathbf{e} = (\mathbf{X}\hat{\boldsymbol{\beta}})'\mathbf{e} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{e} = \hat{\boldsymbol{\beta}}' \cdot \mathbf{0} = 0.$$

Geometrically, $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{e} \perp \mathcal{C}(\mathbf{X})$. □

2.7 Leverages

Definition 2.4 (Leverage). The **leverage** of the i -th observation is h_{ii} , the i -th diagonal entry of $\mathbf{H}$.

Theorem 2.7 (Properties of Leverages).

- (i) $0 \leq h_{ii} \leq 1$ for all i .
- (ii) $\sum_{i=1}^n h_{ii} = k + 1$.

Proof. (i) Since $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H} = \mathbf{H}^2 = \mathbf{H}'\mathbf{H}$. Comparing the (i, i) entry on both sides using $h_{ji} = h_{ij}$:

$$h_{ii} = \sum_{j=1}^n h_{ij}h_{ji} = \sum_{j=1}^n h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2.$$

The rightmost sum is ≥ 0 , so $h_{ii} \geq h_{ii}^2$, which rearranges to $h_{ii}(1 - h_{ii}) \geq 0$. The leftmost expression $h_{ii} = \sum_j h_{ij}^2 \geq 0$ is also a sum of squares. Combining $h_{ii} \geq 0$ with $h_{ii}(1 - h_{ii}) \geq 0$ forces $h_{ii} \leq 1$.

(ii) $\sum_{i=1}^n h_{ii} = \text{tr}(\mathbf{H}) = k + 1$ by Theorem 2.5(iv). □

3 Finite-Sample Properties Without Normality

In this section, only assumptions (A1)–(A4) are used. No distributional assumption on ε is needed.

Before deriving the finite-sample properties of $\hat{\beta}$, we collect the notation for the sum-of-squares and mean-square quantities used throughout the remaining sections.

Definition 3.1 (Sums of Squares and Mean Squares). Given the fitted model $\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}$, define:

- **Total sum of squares:** $\text{SST} = \sum_{i=1}^n (Y_i - \bar{Y})^2$, where $\bar{Y} = n^{-1} \sum_i Y_i$.
- **Regression sum of squares:** $\text{SSR} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$.
- **Error (residual) sum of squares:** $\text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \mathbf{e}'\mathbf{e}$, where $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ is the residual vector of Definition 2.2.
- **Mean squared error:** $\text{MSE} = s^2 = \text{SSE}/(n - k - 1)$, defined when $n > k + 1$.
- **Mean squared regression:** $\text{MSR} = \text{SSR}/k$, defined when $k \geq 1$ (at least one non-intercept regressor).

The decomposition $\text{SST} = \text{SSR} + \text{SSE}$ is proved in Theorem 6.1 (§6); the unbiasedness $\mathbf{E}[\text{MSE}] = \sigma^2$ is proved in Theorem 3.6 below.

We also record a tool used repeatedly in this and later sections.

Lemma 3.1 (Variance and covariance under linear transformations). *Let $\mathbf{z}$ be an $n \times 1$ random vector with mean $\boldsymbol{\mu} = \mathbb{E}[\mathbf{z}]$ and $\text{var}(\mathbf{z}) = \mathbb{E}[(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})']$; for random vectors $\mathbf{U}, \mathbf{V}$ with finite second moments, $\text{cov}(\mathbf{U}, \mathbf{V}) = \mathbb{E}[(\mathbf{U} - \mathbb{E}[\mathbf{U}])(\mathbf{V} - \mathbb{E}[\mathbf{V}])']$. Let $\mathbf{A}$ ($m \times n$), $\mathbf{B}$ ($l \times n$) be constant matrices and $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{d} \in \mathbb{R}^l$ constant vectors. Then*

$$(i) \quad \mathbb{E}[\mathbf{A}\mathbf{z} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b} \text{ and}$$

$$\text{var}(\mathbf{A}\mathbf{z} + \mathbf{b}) = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{A}'; \quad (9)$$

$$(ii) \quad \text{cov}(\mathbf{A}\mathbf{z} + \mathbf{b}, \mathbf{B}\mathbf{z} + \mathbf{d}) = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{B}'.$$

In particular, under assumption (A3), $\text{var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$.

Proof. Let $\boldsymbol{\mu} = \mathbb{E}[\mathbf{z}]$. Since $\mathbf{A}$ and $\mathbf{b}$ are constant, $\mathbb{E}[\mathbf{A}\mathbf{z} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$. Therefore

$$\begin{aligned} \text{var}(\mathbf{A}\mathbf{z} + \mathbf{b}) &= \mathbb{E}[(\mathbf{A}\mathbf{z} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})(\mathbf{A}\mathbf{z} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})'] \\ &= \mathbb{E}[\mathbf{A}(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})' \mathbf{A}'] \\ &= \mathbf{A} \mathbb{E}[(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})'] \mathbf{A}' = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{A}', \end{aligned}$$

since constant matrices pull outside the expectation. Part (ii) is the same computation with $(\mathbf{B}\mathbf{z} + \mathbf{d}) - (\mathbf{B}\boldsymbol{\mu} + \mathbf{d}) = \mathbf{B}(\mathbf{z} - \boldsymbol{\mu})$ in the second factor. Applying (i) to $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ (with $\mathbf{A} = \mathbf{I}_n$, $\mathbf{b} = \mathbf{X}\boldsymbol{\beta}$ constant given the conditioning on $\mathbf{X}$) gives $\text{var}(\mathbf{Y}) = \text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ by (A3). $\square$

3.1 Unbiasedness

Theorem 3.2 (Unbiasedness of $\hat{\beta}$). *Under (A2) and (A4) (the latter is needed for $\hat{\beta}$ to exist), the OLS estimator is unbiased:*

$$\boxed{\mathbb{E}[\hat{\beta}] = \beta.} \quad (10)$$

Proof. Substituting $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$ into $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$:

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \varepsilon) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon \\ &= \mathbf{I}_{k+1}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon. \end{aligned} \quad (11)$$

Taking expectations, using (A2) ($\mathbb{E}[\varepsilon] = \mathbf{0}$):

$$\mathbb{E}[\hat{\beta}] = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\varepsilon] = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \cdot \mathbf{0} = \beta. \quad \square$$

3.2 Variance–Covariance Matrix

Theorem 3.3 (Variance–Covariance Matrix of $\hat{\beta}$). *Under (A3) and (A4):*

$$\boxed{\text{var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.} \quad (12)$$

Proof. From (11), $\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$, which is a linear transformation of ε (with a constant shift). Applying Lemma 3.1 with $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$:

$$\text{var}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \text{var}(\varepsilon) \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}.$$

By (A3), $\text{var}(\varepsilon) = \sigma^2\mathbf{I}_n$. Substituting:

$$\text{var}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1},$$

using $\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{I}_{k+1}$. The diagonal entry $[\text{var}(\hat{\beta})]_{jj} = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$ gives $\text{var}(\hat{\beta}_j)$. □

3.3 The Gauss–Markov Theorem

Theorem 3.4 (Gauss–Markov Theorem). *Under (A1)–(A4), the OLS estimator $\hat{\beta}$ is the **Best Linear Unbiased Estimator** (BLUE) of β . That is, for any other linear unbiased estimator $\tilde{\beta} = \mathbf{C}\mathbf{Y}$ (where $\mathbf{C}$ is a $(k+1) \times n$ constant matrix), the difference $\text{var}(\tilde{\beta}) - \text{var}(\hat{\beta})$ is positive semi-definite. In particular, $\text{var}(\tilde{\beta}_j) \geq \text{var}(\hat{\beta}_j)$ for every j .*

Proof. **Step 1: Necessary condition for unbiasedness.** By (A1), $E[\tilde{\beta}] = \mathbf{C}\mathbf{X}\beta = \beta$ for all $\beta \in \mathbb{R}^{k+1}$ forces $\mathbf{C}\mathbf{X} = \mathbf{I}_{k+1}$ (setting $\mathbf{v} = \mathbf{e}_j$ in $\mathbf{C}\mathbf{X}\mathbf{v} = \mathbf{v}$ reads off the j -th column).

Step 2: Decompose $\mathbf{C}$. Write $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$, where $\mathbf{D}$ is the excess beyond the OLS coefficient matrix. From $\mathbf{C}\mathbf{X} = \mathbf{I}_{k+1}$:

$$\mathbf{D}\mathbf{X} = \mathbf{C}\mathbf{X} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{I}_{k+1} - \mathbf{I}_{k+1} = \mathbf{0}.$$

Step 3: Compute $\text{var}(\tilde{\beta})$. Since $\tilde{\beta} = \mathbf{C}\mathbf{Y}$ with $\mathbf{C}$ constant, Lemma 3.1 gives $\text{var}(\tilde{\beta}) = \mathbf{C} \text{var}(\mathbf{Y}) \mathbf{C}' = \sigma^2 \mathbf{C}\mathbf{C}'$:

$$\text{var}(\tilde{\beta}) = \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}] [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'].$$

Expanding yields four terms:

$$\begin{aligned}
 \sigma^2 \mathbf{C}\mathbf{C}' &= \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} && \text{(term A)} \\
 &+ \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{D}' && \text{(term B)} \\
 &+ \sigma^2 \mathbf{D}\mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} && \text{(term C)} \\
 &+ \sigma^2 \mathbf{D}\mathbf{D}'. && \text{(term D)}
 \end{aligned}$$

Term A simplifies to $\sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$ because $\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{I}_{k+1}$. Term C vanishes because $\mathbf{D}\mathbf{X} = \mathbf{0}$ (Step 2). Term B equals (term C)', so it also vanishes. Hence:

$$\text{var}(\tilde{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} + \sigma^2 \mathbf{D}\mathbf{D}' = \text{var}(\hat{\boldsymbol{\beta}}) + \sigma^2 \mathbf{D}\mathbf{D}'.$$

Since $\mathbf{D}\mathbf{D}'$ is positive semi-definite (for any $\mathbf{v} \in \mathbb{R}^{k+1}$, $\mathbf{v}'\mathbf{D}\mathbf{D}'\mathbf{v} = \|\mathbf{D}'\mathbf{v}\|^2 \geq 0$):

$$\text{var}(\tilde{\boldsymbol{\beta}}) - \text{var}(\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{D}\mathbf{D}' \succeq \mathbf{0}.$$

Equality holds if and only if $\mathbf{D}'\mathbf{v} = \mathbf{0}$ for all $\mathbf{v}$, i.e., $\mathbf{D} = \mathbf{0}$; equivalently, $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$. □

3.4 Estimation of σ^2

Lemma 3.5 (Expected Value of a Quadratic Form). *Let $\mathbf{z}$ be an $n \times 1$ random vector with $\mathbb{E}[\mathbf{z}] = \boldsymbol{\mu}$ and $\text{var}(\mathbf{z}) = \boldsymbol{\Sigma}$. Let $\mathbf{A}$ be an $n \times n$ constant matrix. Then:*

$$\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}. \quad (13)$$

In particular, if $\mathbb{E}[\mathbf{z}] = \mathbf{0}$, then $\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$.

Proof. The quadratic form is a scalar: $\mathbf{z}'\mathbf{A}\mathbf{z} = \sum_{i,j} a_{ij}z_iz_j$. Taking expectations:

$$\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \sum_{i,j} a_{ij}\mathbb{E}[z_iz_j].$$

From $\text{cov}(z_i, z_j) = \mathbb{E}[z_iz_j] - \mu_i\mu_j$, we get $\mathbb{E}[z_iz_j] = \Sigma_{ij} + \mu_i\mu_j$. Substituting:

$$\mathbb{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \sum_{i,j} a_{ij}\Sigma_{ij} + \sum_{i,j} a_{ij}\mu_i\mu_j.$$

The first sum equals $\text{tr}(\mathbf{A}\boldsymbol{\Sigma})$: indeed, $\text{tr}(\mathbf{A}\boldsymbol{\Sigma}) = \sum_i (\mathbf{A}\boldsymbol{\Sigma})_{ii} = \sum_i \sum_j a_{ij}\Sigma_{ji} = \sum_{i,j} a_{ij}\Sigma_{ij}$, where the last step uses the symmetry $\Sigma_{ji} = \Sigma_{ij}$. The second sum equals $\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$: indeed, $\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu} = \sum_i \mu_i (\mathbf{A}\boldsymbol{\mu})_i = \sum_i \mu_i \sum_j a_{ij}\mu_j = \sum_{i,j} a_{ij}\mu_i\mu_j$. Combining yields (13). $\square$

Theorem 3.6 (Unbiasedness of MSE). *Under (A2)–(A4), the **mean squared error***

$$s^2 = \text{MSE} = \frac{\text{SSE}}{n - k - 1} = \frac{\mathbf{e}'\mathbf{e}}{n - k - 1} \quad (14)$$

is an unbiased estimator of σ^2 : $\text{E}[\text{MSE}] = \sigma^2$.

Proof. Step 1: Express SSE as a quadratic form in $\boldsymbol{\varepsilon}$. Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ (Remark after Definition 2.2):

$$\text{SSE} = \mathbf{e}'\mathbf{e} = (\mathbf{M}\boldsymbol{\varepsilon})'(\mathbf{M}\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}'\mathbf{M}'\mathbf{M}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon},$$

where the last step uses $\mathbf{M}'\mathbf{M} = \mathbf{M}\mathbf{M} = \mathbf{M}$ (symmetric idempotent).

Step 2: Apply Lemma 3.5. With $\mathbf{z} = \boldsymbol{\varepsilon}$, $\mathbf{A} = \mathbf{M}$, $\boldsymbol{\mu} = \text{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (by (A2)), and $\boldsymbol{\Sigma} = \text{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ (by (A3)):

$$\text{E}[\text{SSE}] = \text{E}[\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}] = \text{tr}(\mathbf{M} \cdot \sigma^2\mathbf{I}) + \mathbf{0}'\mathbf{M}\mathbf{0} = \sigma^2 \text{tr}(\mathbf{M}).$$

Step 3: Evaluate the trace. By Theorem 2.5(iv), $\text{tr}(\mathbf{M}) = n - k - 1$. Hence $\text{E}[\text{SSE}] = \sigma^2(n - k - 1)$.

Step 4: Conclude.

$$\text{E}[\text{MSE}] = \text{E}\left[\frac{\text{SSE}}{n - k - 1}\right] = \frac{\sigma^2(n - k - 1)}{n - k - 1} = \sigma^2. \quad \square$$

3.5 Variance of Individual Residuals

Theorem 3.7 (Variance of Residuals). *Under (A3) and (A4):*

$$\text{var}(\mathbf{e}) = \sigma^2 \mathbf{M}, \quad \text{and in particular, } \text{var}(e_i) = \sigma^2(1 - h_{ii}). \quad (15)$$

Proof. Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ and $\mathbf{M}$ is symmetric ($\mathbf{M}' = \mathbf{M}$) and idempotent ($\mathbf{M}^2 = \mathbf{M}$):

$$\text{var}(\mathbf{e}) = \mathbf{M} \text{var}(\boldsymbol{\varepsilon}) \mathbf{M}' = \mathbf{M}(\sigma^2 \mathbf{I}) \mathbf{M}' = \sigma^2 \mathbf{M} \mathbf{M} = \sigma^2 \mathbf{M}^2 = \sigma^2 \mathbf{M}.$$

The (i, i) entry gives $\text{var}(e_i) = \sigma^2 m_{ii}$. Since $\mathbf{M} = \mathbf{I} - \mathbf{H}$, we have $m_{ii} = 1 - h_{ii}$, so $\text{var}(e_i) = \sigma^2(1 - h_{ii})$. $\square$

Remark. By Theorem 2.7(i), $0 \leq h_{ii} \leq 1$, so $\text{var}(e_i) \leq \sigma^2$. High-leverage observations (large h_{ii}) produce residuals with smaller variance: the fitted line is pulled toward them, leaving less room for the residual to vary.

4 Multivariate Normal Tools

This section collects the distributional results needed for exact inference in Section 5.

Theorem 4.1 (Closure Under Linear Transformations). *If $\mathbf{z} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is an $n \times 1$ normal random vector, $\mathbf{A}$ is an $m \times n$ constant matrix, and $\mathbf{b} \in \mathbb{R}^m$ is a constant vector, then*

$$\mathbf{A}\mathbf{z} + \mathbf{b} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'). \quad (16)$$

Proof. The moment-generating function (MGF) of $\mathbf{z}$ is $M_{\mathbf{z}}(\mathbf{t}) = \exp(\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t})$. The MGF of $\mathbf{w} = \mathbf{A}\mathbf{z} + \mathbf{b}$ is:

$$\begin{aligned} M_{\mathbf{w}}(\mathbf{t}) &= \mathbb{E}\left[e^{\mathbf{t}'(\mathbf{A}\mathbf{z}+\mathbf{b})}\right] = e^{\mathbf{t}'\mathbf{b}} \cdot \mathbb{E}\left[e^{(\mathbf{A}'\mathbf{t})'\mathbf{z}}\right] = e^{\mathbf{t}'\mathbf{b}} \cdot M_{\mathbf{z}}(\mathbf{A}'\mathbf{t}) \\ &= e^{\mathbf{t}'\mathbf{b}} \cdot \exp((\mathbf{A}'\mathbf{t})'\boldsymbol{\mu} + \frac{1}{2}(\mathbf{A}'\mathbf{t})'\boldsymbol{\Sigma}(\mathbf{A}'\mathbf{t})) \\ &= \exp(\mathbf{t}'(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2}\mathbf{t}'\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'\mathbf{t}). \end{aligned}$$

This is the MGF of $N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$. By uniqueness of the MGF, the result follows. $\square$

Theorem 4.2 (Uncorrelatedness $\Leftrightarrow$ Independence Under Joint Normality). *Let $(\mathbf{U}', \mathbf{V}')$ be jointly normal:*

$$\begin{pmatrix} \mathbf{U} \\ \mathbf{V} \end{pmatrix} \sim N\left(\begin{pmatrix} \boldsymbol{\mu}_U \\ \boldsymbol{\mu}_V \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{UU} & \boldsymbol{\Sigma}_{UV} \\ \boldsymbol{\Sigma}_{VU} & \boldsymbol{\Sigma}_{VV} \end{pmatrix}\right).$$

Then $\boldsymbol{\Sigma}_{UV} = \mathbf{0}$ if and only if $\mathbf{U}$ and $\mathbf{V}$ are independent.

Proof. ($\Leftarrow$): Independence implies uncorrelatedness for any distribution.

($\Rightarrow$): With $\boldsymbol{\Sigma}_{UV} = \mathbf{0}$, the covariance is block-diagonal. The joint MGF is $M_{(\mathbf{U}, \mathbf{V})}(\mathbf{s}, \mathbf{t}) = \exp((\mathbf{s}', \mathbf{t}')\boldsymbol{\mu} +$

$\frac{1}{2}(\mathbf{s}', \mathbf{t}')\boldsymbol{\Sigma}(\mathbf{s}', \mathbf{t}')'$. Block expansion gives

$$(\mathbf{s}', \mathbf{t}')\boldsymbol{\mu} = \mathbf{s}'\boldsymbol{\mu}_U + \mathbf{t}'\boldsymbol{\mu}_V, \quad (\mathbf{s}', \mathbf{t}')\boldsymbol{\Sigma} \begin{pmatrix} \mathbf{s} \\ \mathbf{t} \end{pmatrix} = \mathbf{s}'\boldsymbol{\Sigma}_{UU}\mathbf{s} + \mathbf{t}'\boldsymbol{\Sigma}_{VV}\mathbf{t},$$

using $\boldsymbol{\Sigma}_{UV} = \boldsymbol{\Sigma}'_{VU} = \mathbf{0}$. The joint MGF factors:

$$M_{(\mathbf{U}, \mathbf{V})}(\mathbf{s}, \mathbf{t}) = \exp(\mathbf{s}'\boldsymbol{\mu}_U + \frac{1}{2}\mathbf{s}'\boldsymbol{\Sigma}_{UU}\mathbf{s}) \cdot \exp(\mathbf{t}'\boldsymbol{\mu}_V + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}_{VV}\mathbf{t}) = M_{\mathbf{U}}(\mathbf{s})M_{\mathbf{V}}(\mathbf{t}),$$

which implies independence. □

Remark. For general (non-normal) distributions, uncorrelatedness does *not* imply independence. The equivalence is a special property of the multivariate normal.

Theorem 4.3 (Chi-Square Distribution of Quadratic Forms). *Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{A}$ be an $n \times n$ symmetric idempotent matrix with $\mathbf{A} \neq \mathbf{0}$ (so that $r = \text{rank}(\mathbf{A}) \geq 1$, as the definition of χ_r^2 requires). Then*

$$\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi_r^2, \quad \text{where } r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}). \quad (17)$$

Proof. Step 1: Spectral decomposition. Since $\mathbf{A}$ is symmetric, there exists an orthogonal matrix $\mathbf{P}$ ($\mathbf{P}'\mathbf{P} = \mathbf{P}\mathbf{P}' = \mathbf{I}$) such that $\mathbf{A} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}'$, where $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$. By idempotency and Lemma 2.4, each $\lambda_i \in \{0, 1\}$.

Without loss of generality, let the first r eigenvalues be 1 and the rest be 0:

$$\mathbf{\Lambda} = \text{diag}(\underbrace{1, \dots, 1}_r, \underbrace{0, \dots, 0}_{n-r}).$$

Step 2: Apply orthogonal transformation. Let $\mathbf{w} = \mathbf{P}'\mathbf{z}$. Since $\mathbf{P}$ is orthogonal:

$$\mathbf{E}[\mathbf{w}] = \mathbf{P}'\mathbf{E}[\mathbf{z}] = \mathbf{0}, \quad \text{var}(\mathbf{w}) = \mathbf{P}'\mathbf{I}\mathbf{P} = \mathbf{I}_n.$$

By Theorem 4.1, $\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_n)$, so $w_1, \dots, w_n$ are i.i.d. $N(0, 1)$.

Step 3: Evaluate the quadratic form.

$$\mathbf{z}'\mathbf{A}\mathbf{z} = \mathbf{z}'\mathbf{P}\mathbf{\Lambda}\mathbf{P}'\mathbf{z} = (\mathbf{P}'\mathbf{z})'\mathbf{\Lambda}(\mathbf{P}'\mathbf{z}) = \mathbf{w}'\mathbf{\Lambda}\mathbf{w} = \sum_{i=1}^n \lambda_i w_i^2 = \sum_{i=1}^r w_i^2.$$

Step 4: Conclude. Since $w_1, \dots, w_r$ are independent $N(0, 1)$, by definition $\sum_{i=1}^r w_i^2 \sim \chi_r^2$. □

Theorem 4.4 (Independence of Quadratic Forms — a Special Case of Craig's Theorem). *Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$, and let $\mathbf{A}_1, \mathbf{A}_2$ be $n \times n$ symmetric idempotent matrices with $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$. Then $\mathbf{z}'\mathbf{A}_1\mathbf{z}$ and $\mathbf{z}'\mathbf{A}_2\mathbf{z}$ are independent.*

Proof. Define $\mathbf{U} = \mathbf{A}_1 \mathbf{z}$ and $\mathbf{V} = \mathbf{A}_2 \mathbf{z}$. Stack them into a single vector obtained from $\mathbf{z}$ by one linear map:

$$\begin{pmatrix} \mathbf{U} \\ \mathbf{V} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{pmatrix} \mathbf{z}.$$

Since $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$, Theorem 4.1 implies $(\mathbf{U}', \mathbf{V}')'$ is *jointly* normal. Their cross-covariance is, by Lemma 3.1(ii),

$$\text{cov}(\mathbf{U}, \mathbf{V}) = \mathbf{A}_1 \text{var}(\mathbf{z}) \mathbf{A}_2' = \mathbf{A}_1 \mathbf{I} \mathbf{A}_2 = \mathbf{A}_1 \mathbf{A}_2 = \mathbf{0},$$

where $\mathbf{A}_2' = \mathbf{A}_2$ by symmetry. By Theorem 4.2, $\mathbf{U}$ and $\mathbf{V}$ are independent. By symmetric idempotency of $\mathbf{A}_1$, $\mathbf{z}' \mathbf{A}_1 \mathbf{z} = \mathbf{z}' \mathbf{A}_1' \mathbf{A}_1 \mathbf{z} = \|\mathbf{U}\|^2$ is a function of $\mathbf{U}$ alone; similarly $\mathbf{z}' \mathbf{A}_2 \mathbf{z} = \|\mathbf{V}\|^2$ depends only on $\mathbf{V}$. The two quadratic forms are therefore independent. $\square$

Remark. In Lemma 7.2 (Section 7), Theorem 4.4 is applied with $\mathbf{A}_1 = \mathbf{H} - n^{-1} \mathbf{1} \mathbf{1}'$ and $\mathbf{A}_2 = \mathbf{M}$ to prove $\text{SSR} \perp\!\!\!\perp \text{SSE}$.

5 Exact Inference Under Normality

From this section onward, all five assumptions (A1)–(A5) are in force.

5.1 Sampling Distribution of $\hat{\beta}$

Theorem 5.1 (Normality of $\hat{\beta}$). *Under (A1)–(A5):*

$$\hat{\beta} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}). \quad (18)$$

Proof. From (11), $\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$. This is an affine transformation of ε . Under (A5), $\varepsilon \sim N(\mathbf{0}, \sigma^2\mathbf{I})$. By Theorem 4.1 with $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, $\mathbf{z} = \varepsilon$, $\mathbf{b} = \beta$:

$$\begin{aligned} \hat{\beta} &\sim N(\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \cdot \mathbf{0}, (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}) \\ &= N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}). \end{aligned}$$

□

5.2 Distribution of SSE/σ^2 and Independence from $\hat{\beta}$

Theorem 5.2 (Chi-Square Distribution of SSE/σ^2). *Under (A1)–(A5):*

$$\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-k-1}^2, \quad (19)$$

and SSE is independent of $\hat{\beta}$.

Proof. **Step 1: Standardize.** Let $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma$. Under (A5), $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$. Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$:

$$\frac{\text{SSE}}{\sigma^2} = \frac{\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}}{\sigma^2} = \mathbf{z}'\mathbf{M}\mathbf{z}.$$

Step 2: Apply Theorem 4.3. The matrix $\mathbf{M}$ is symmetric and idempotent (Theorem 2.5(i)), with $\text{tr}(\mathbf{M}) = n - k - 1$ (Theorem 2.5(iv)). Hence $\mathbf{z}'\mathbf{M}\mathbf{z} \sim \chi_{n-k-1}^2$.

Step 3: Prove independence. Stack $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ and $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ into a single vector obtained from $\boldsymbol{\varepsilon}$ by one linear map:

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ \mathbf{M} \end{pmatrix} \boldsymbol{\varepsilon}.$$

Since $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, Theorem 4.1 makes this stacked vector *jointly* normal (individual normality alone would not suffice; joint normality requires that both pieces come from a single normal vector). Their cross-covariance is:

$$\begin{aligned} \text{cov}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}, \mathbf{M}\boldsymbol{\varepsilon}) &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \text{var}(\boldsymbol{\varepsilon})\mathbf{M}' \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{M} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0}, \end{aligned}$$

using $\mathbf{X}'\mathbf{M} = (\mathbf{M}\mathbf{X})' = \mathbf{0}$ (Theorem 2.5(ii)). By Theorem 4.2, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ and $\mathbf{e}$ are independent. Since $\hat{\boldsymbol{\beta}}$ is $(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ plus a constant and $\text{SSE} = \mathbf{e}'\mathbf{e}$ is a function of $\mathbf{e}$, $\hat{\boldsymbol{\beta}}$ is independent of $\mathbf{e}$ (hence of SSE). $\square$

5.3 OLS as Maximum Likelihood Under Normality

Theorem 5.3 (OLS Equals MLE Under Normality). *Under (A1)–(A5), the OLS estimator $\hat{\beta}$ is also the maximum likelihood estimator of β . The MLE of σ^2 is $\hat{\sigma}_{\text{MLE}}^2 = \text{SSE}/n$, which is biased.*

Proof. Under (A5), $\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$, so by Theorem 4.1 applied to the affine transformation $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$, $\mathbf{Y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I})$. The log-likelihood is:

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta).$$

Maximizing over β : In $\ell(\beta, \sigma^2)$, the first two terms do not involve β , and the last term is $-(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)/(2\sigma^2)$ with $\sigma^2 > 0$ treated as fixed. Since $-1/(2\sigma^2) < 0$, maximizing ℓ over β is equivalent to minimizing $(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) = Q(\beta)$. This is exactly the OLS problem, so $\hat{\beta}_{\text{MLE}} = \hat{\beta}_{\text{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$.

Maximizing over σ^2 : Substituting $\hat{\beta}$ and noting that $(\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta}) = \text{SSE}$, the profile log-likelihood becomes

$$\ell(\hat{\beta}, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{\text{SSE}}{2\sigma^2}.$$

Differentiating with respect to σ^2 :

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\text{SSE}}{2(\sigma^2)^2} = 0 \implies \hat{\sigma}_{\text{MLE}}^2 = \frac{\text{SSE}}{n}.$$

Second-order condition: Writing $u = \sigma^2$ for brevity,

$$\frac{\partial^2 \ell}{\partial u^2} = \frac{n}{2u^2} - \frac{\text{SSE}}{u^3}.$$

At the critical point $\hat{u} = \text{SSE}/n$, so $u^2 = \text{SSE}^2/n^2$ and $u^3 = \text{SSE}^3/n^3$:

$$\left. \frac{\partial^2 \ell}{\partial u^2} \right|_{\hat{u}} = \frac{n \cdot n^2}{2\text{SSE}^2} - \frac{\text{SSE} \cdot n^3}{\text{SSE}^3} = \frac{n^3}{2\text{SSE}^2} - \frac{n^3}{\text{SSE}^2} = -\frac{n^3}{2\text{SSE}^2} < 0,$$

confirming a maximum.

Bias of $\hat{\sigma}_{\text{MLE}}^2$. Using Theorem 3.6 ($\mathbb{E}[\text{SSE}] = (n - k - 1)\sigma^2$):

$$\mathbb{E}[\hat{\sigma}_{\text{MLE}}^2] = \mathbb{E}\left[\frac{\text{SSE}}{n}\right] = \frac{(n - k - 1)\sigma^2}{n} = \frac{n - k - 1}{n}\sigma^2 < \sigma^2,$$

so $\hat{\sigma}_{\text{MLE}}^2$ underestimates σ^2 on average by a factor of $(n - k - 1)/n$. The unbiased estimator is $\text{MSE} = \text{SSE}/(n - k - 1)$, which rescales by this factor. $\square$

6 Sum of Squares and the Coefficient of Determination

6.1 Variance Decomposition

Definition 6.1 (Centering Matrix). Let $\mathbf{J} = \frac{1}{n}\mathbf{1}\mathbf{1}' = \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}'$, the hat matrix (Definition 2.2) of the intercept-only design $\mathbf{X} = \mathbf{1}$; by Remark 2.4 it is the orthogonal projection onto $\mathcal{C}(\mathbf{1})$, and $\mathbf{J}\mathbf{v} = \bar{v}\mathbf{1}$ for any vector $\mathbf{v}$, where $\bar{v} = \frac{1}{n}\mathbf{1}'\mathbf{v}$. The **centering matrix** is the corresponding residual maker $\mathbf{M}_0 = \mathbf{I}_n - \mathbf{J} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$. For any vector $\mathbf{v}$, $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1}$.

Remark (Notation: $\mathbf{M}_0$ versus $\mathbf{M}$). The subscript “0” in $\mathbf{M}_0$ signals that it is the residual maker with respect to the *intercept only* (the single column $\mathbf{1}$), whereas $\mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ from Definition 2.2 is the residual maker with respect to the *full* design matrix $\mathbf{X}$. Conceptually,

$\mathbf{M}_0$ centers a vector, $\mathbf{M}$ regresses out all of $\mathbf{X}$.

Both are symmetric idempotent, but they project onto different subspaces:

$$\mathbf{M}_0 : \mathbb{R}^n \rightarrow \mathbf{1}^\perp, \quad \mathbf{M} : \mathbb{R}^n \rightarrow \mathcal{C}(\mathbf{X})^\perp.$$

If the model includes an intercept then $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, hence $\mathcal{C}(\mathbf{X})^\perp \subseteq \mathbf{1}^\perp$, and so $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0\mathbf{M} = \mathbf{M}$ (this follows from the projection identity $\mathbf{H}\mathbf{J} = \mathbf{J}\mathbf{H} = \mathbf{J}$ proved in Lemma 7.2, Step 2, by taking complements). In particular, $\mathbf{M}$ and $\mathbf{M}_0$ are not equal, nor is one a simple scalar multiple of the other; they coincide only in the trivial case $\mathbf{X} = \mathbf{1}$.

Remark. $\mathbf{M}_0$ is symmetric and idempotent with $\text{tr}(\mathbf{M}_0) = n - 1$: it is the residual maker $\mathbf{M}$ of the intercept-only design $\mathbf{X} = \mathbf{1}$, so Theorem 2.5(i) and (iv) apply with $k = 0$.

Theorem 6.1 (Variance Decomposition). *In the model with an intercept, the sums of squares introduced in Definition 3.1 satisfy:*

$$\boxed{\text{SST} = \text{SSR} + \text{SSE}}, \quad (20)$$

with matrix expressions:

- $\text{SST} = \sum (Y_i - \bar{Y})^2 = \mathbf{Y}'\mathbf{M}_0\mathbf{Y}$ (Total Sum of Squares),
- $\text{SSR} = \sum (\hat{Y}_i - \bar{Y})^2 = \hat{\mathbf{Y}}'\mathbf{M}_0\hat{\mathbf{Y}}$ (Regression Sum of Squares),
- $\text{SSE} = \sum (Y_i - \hat{Y}_i)^2 = \mathbf{e}'\mathbf{e}$ (Error/Residual Sum of Squares).

Proof. Step 1: Verify the matrix expressions for SST and SSR. For any vector $\mathbf{v} \in \mathbb{R}^n$, Definition 6.1 gives $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1}$, so

$$\mathbf{v}'\mathbf{M}_0\mathbf{v} = (\mathbf{M}_0\mathbf{v})'(\mathbf{M}_0\mathbf{v}) = \|\mathbf{v} - \bar{v}\mathbf{1}\|^2 = \sum_{i=1}^n (v_i - \bar{v})^2,$$

using $\mathbf{M}_0'\mathbf{M}_0 = \mathbf{M}_0^2 = \mathbf{M}_0$ (symmetric idempotent). Applying this with $\mathbf{v} = \mathbf{Y}$ gives $\text{SST} = \mathbf{Y}'\mathbf{M}_0\mathbf{Y}$; with $\mathbf{v} = \hat{\mathbf{Y}}$ gives $\text{SSR} = \hat{\mathbf{Y}}'\mathbf{M}_0\hat{\mathbf{Y}}$, provided $\overline{\hat{Y}} = \bar{Y}$ —which follows from $\mathbf{1}'\mathbf{e} = 0$ (Theorem 2.6(ii)) since $\mathbf{1}'\hat{\mathbf{Y}} = \mathbf{1}'(\mathbf{Y} - \mathbf{e}) = \mathbf{1}'\mathbf{Y}$.

Step 2: Decompose $\mathbf{M}_0\mathbf{Y}$ into orthogonal components. Write

$$\mathbf{M}_0\mathbf{Y} = \mathbf{M}_0(\hat{\mathbf{Y}} + \mathbf{e}) = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{M}_0\mathbf{e} = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e}, \quad (21)$$

where the last equality uses $\mathbf{M}_0\mathbf{e} = \mathbf{e} - \bar{e}\mathbf{1} = \mathbf{e}$ because $\bar{e} = n^{-1}\mathbf{1}'\mathbf{e} = 0$ (Theorem 2.6(ii)).

Step 3: Expand SST. Using Step 2 and $\mathbf{M}_0^2 = \mathbf{M}_0$:

$$\begin{aligned} \text{SST} &= \mathbf{Y}'\mathbf{M}_0\mathbf{Y} = (\mathbf{M}_0\mathbf{Y})'(\mathbf{M}_0\mathbf{Y}) = (\mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e})'(\mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e}) \\ &= \text{SSR} + 2\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} + \text{SSE}. \end{aligned}$$

Step 4: The cross term vanishes. Using $\mathbf{e}'\hat{\mathbf{Y}} = 0$ and $\mathbf{e}'\mathbf{1} = 0$ (Theorem 2.6(ii), (iii)):

$$\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} = \mathbf{e}'\hat{\mathbf{Y}} - \bar{Y}\mathbf{e}'\mathbf{1} = 0.$$

Hence $\text{SST} = \text{SSR} + \text{SSE}$. □

Remark (Geometric view). Equation (21) is an orthogonal decomposition of $\mathbf{M}_0\mathbf{Y}$ into two pieces: the centered fitted values $\mathbf{M}_0\hat{\mathbf{Y}}$, which lie in the subspace $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$ (the non-intercept directions of the column space), and the residuals $\mathbf{e}$, which lie in $\mathcal{C}(\mathbf{X})^\perp$. Pythagoras applied to these two orthogonal pieces is precisely $\text{SST} = \text{SSR} + \text{SSE}$.

6.2 Coefficient of Determination

Definition 6.2 (R^2 and Adjusted R^2). Assume the model has an intercept and $SST > 0$, i.e. $\mathbf{Y}$ is not a constant vector ($SST = \|\mathbf{M}_0 \mathbf{Y}\|^2 = 0$ iff $\mathbf{Y} = \bar{Y} \mathbf{1}$). The **coefficient of determination** is

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}. \quad (22)$$

It measures the proportion of total variation in $\mathbf{Y}$ explained by the regression. The **adjusted** R^2 is

$$\bar{R}^2 = 1 - \frac{SSE/(n - k - 1)}{SST/(n - 1)} = 1 - \frac{n - 1}{n - k - 1}(1 - R^2) = 1 - \frac{MSE}{s_Y^2}, \quad (23)$$

where $s_Y^2 = SST/(n - 1)$. The adjusted R^2 can be negative.

Proposition 6.2 (Bounds on R^2). *In any model with an intercept and $SST > 0$, $0 \leq R^2 \leq 1$.*

Proof. $R^2 = SSR/SST \geq 0$ because $SSR = \sum(\hat{Y}_i - \bar{Y})^2 \geq 0$ and $SST = \sum(Y_i - \bar{Y})^2 > 0$ by hypothesis. For the upper bound, Theorem 6.1 gives $SST = SSR + SSE$ with $SSE \geq 0$, so $SSR \leq SST$, i.e., $R^2 \leq 1$. $\square$

Theorem 6.3 (Monotonicity of R^2). *Adding a regressor to the model cannot decrease R^2 . That is, if R_k^2 is the coefficient of determination with k regressors and R_{k+1}^2 is that with an additional regressor, then $R_{k+1}^2 \geq R_k^2$.*

Proof. Let $\mathcal{C}_k \subseteq \mathcal{C}_{k+1}$ denote the column spaces of the smaller and larger design matrices (the inclusion holds because every column of $\mathbf{X}_k$ is also a column of $\mathbf{X}_{k+1}$). Write $\hat{\mathbf{Y}}_k, \hat{\mathbf{Y}}_{k+1}$ for the corresponding OLS fits. By Theorem 2.3, $\hat{\mathbf{Y}}_{k+1}$ is the point in $\mathcal{C}_{k+1}$ closest to $\mathbf{Y}$, so $\|\mathbf{Y} - \hat{\mathbf{Y}}_{k+1}\| \leq \|\mathbf{Y} - \mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{C}_{k+1}$. In particular, $\hat{\mathbf{Y}}_k \in \mathcal{C}_k \subseteq \mathcal{C}_{k+1}$, so

$$\text{SSE}_{k+1} = \|\mathbf{Y} - \hat{\mathbf{Y}}_{k+1}\|^2 \leq \|\mathbf{Y} - \hat{\mathbf{Y}}_k\|^2 = \text{SSE}_k.$$

Since SST does not depend on the regressors,

$$R_{k+1}^2 = 1 - \frac{\text{SSE}_{k+1}}{\text{SST}} \geq 1 - \frac{\text{SSE}_k}{\text{SST}} = R_k^2. \quad \square$$

Lemma 6.4 (Reduction in SSE from adding one regressor). *Write the enlarged design as $\mathbf{X}_{k+1} = [\mathbf{X}_k \ \mathbf{x}]$, where $\mathbf{x}$ is the new regressor column, and assume $\mathbf{x}'\mathbf{M}_k\mathbf{x} > 0$ (where $\mathbf{M}_k$ is the residual maker for $\mathbf{X}_k$), which holds whenever $\mathbf{x}$ is not in $\mathcal{C}(\mathbf{X}_k)$. Then the OLS coefficient on $\mathbf{x}$ in the enlarged model is*

$$\hat{\beta}_{k+1} = \frac{\mathbf{x}'\mathbf{M}_k\mathbf{Y}}{\mathbf{x}'\mathbf{M}_k\mathbf{x}}, \quad (24)$$

and the reduction in residual sum of squares is

$$\text{SSE}_k - \text{SSE}_{k+1} = \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x} = \frac{(\mathbf{x}'\mathbf{M}_k\mathbf{Y})^2}{\mathbf{x}'\mathbf{M}_k\mathbf{x}} \geq 0. \quad (25)$$

This identity uses only normal equations (no forward dependence on Section 8).

Proof. The partitioned normal equations $\mathbf{X}'_{k+1}\mathbf{X}_{k+1}\hat{\boldsymbol{\beta}}_{k+1} = \mathbf{X}'_{k+1}\mathbf{Y}$ read

$$\mathbf{X}'_k\mathbf{X}_k\hat{\boldsymbol{\beta}}_k^* + \mathbf{X}'_k\mathbf{x}\hat{\beta}_{k+1} = \mathbf{X}'_k\mathbf{Y}, \quad (\text{a})$$

$$\mathbf{x}'\mathbf{X}_k\hat{\boldsymbol{\beta}}_k^* + \mathbf{x}'\mathbf{x}\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{Y}, \quad (\text{b})$$

where $\hat{\boldsymbol{\beta}}_k^*$ denotes the first- k block of the enlarged estimator (generally different from the short-regression

estimator $\hat{\beta}_k$). From (a), $\hat{\beta}_k^* = (\mathbf{X}'_k \mathbf{X}_k)^{-1} \mathbf{X}'_k (\mathbf{Y} - \mathbf{x} \hat{\beta}_{k+1})$. Substituting into (b):

$$\mathbf{x}' \mathbf{X}_k (\mathbf{X}'_k \mathbf{X}_k)^{-1} \mathbf{X}'_k (\mathbf{Y} - \mathbf{x} \hat{\beta}_{k+1}) + \mathbf{x}' \mathbf{x} \hat{\beta}_{k+1} = \mathbf{x}' \mathbf{Y}.$$

Writing $\mathbf{H}_k = \mathbf{X}_k (\mathbf{X}'_k \mathbf{X}_k)^{-1} \mathbf{X}'_k = \mathbf{I} - \mathbf{M}_k$ and rearranging:

$$\mathbf{x}' \mathbf{H}_k \mathbf{Y} - \mathbf{x}' \mathbf{H}_k \mathbf{x} \hat{\beta}_{k+1} + \mathbf{x}' \mathbf{x} \hat{\beta}_{k+1} = \mathbf{x}' \mathbf{Y} \iff \mathbf{x}' (\mathbf{I} - \mathbf{H}_k) \mathbf{x} \hat{\beta}_{k+1} = \mathbf{x}' (\mathbf{I} - \mathbf{H}_k) \mathbf{Y},$$

which gives (24).

For (25), relate $\hat{\beta}_k^*$ to the short-regression estimator $\hat{\beta}_k = (\mathbf{X}'_k \mathbf{X}_k)^{-1} \mathbf{X}'_k \mathbf{Y}$: from (a),

$$\hat{\beta}_k^* = \hat{\beta}_k - (\mathbf{X}'_k \mathbf{X}_k)^{-1} \mathbf{X}'_k \mathbf{x} \hat{\beta}_{k+1}.$$

Multiplying by $\mathbf{X}_k$ gives $\mathbf{X}_k \hat{\beta}_k^* = \hat{\mathbf{Y}}_k - \mathbf{H}_k \mathbf{x} \hat{\beta}_{k+1}$, so

$$\hat{\mathbf{Y}}_{k+1} = \mathbf{X}_k \hat{\beta}_k^* + \mathbf{x} \hat{\beta}_{k+1} = \hat{\mathbf{Y}}_k + (\mathbf{I} - \mathbf{H}_k) \mathbf{x} \hat{\beta}_{k+1} = \hat{\mathbf{Y}}_k + \mathbf{M}_k \mathbf{x} \hat{\beta}_{k+1},$$

and the residuals satisfy $\mathbf{e}_{k+1} = \mathbf{e}_k - \mathbf{M}_k \mathbf{x} \hat{\beta}_{k+1}$. Taking squared norms, using $\mathbf{e}_k = \mathbf{M}_k \mathbf{Y}$ and $\mathbf{M}_k^2 = \mathbf{M}_k$:

$$\text{SSE}_{k+1} = \text{SSE}_k - 2\hat{\beta}_{k+1} \cdot \mathbf{Y}' \mathbf{M}_k \mathbf{x} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}' \mathbf{M}_k \mathbf{x}.$$

Substituting $\hat{\beta}_{k+1} = (\mathbf{x}' \mathbf{M}_k \mathbf{Y}) / (\mathbf{x}' \mathbf{M}_k \mathbf{x})$ makes the middle term $-2\hat{\beta}_{k+1}^2 \cdot \mathbf{x}' \mathbf{M}_k \mathbf{x}$, so

$$\text{SSE}_{k+1} = \text{SSE}_k - \hat{\beta}_{k+1}^2 \cdot \mathbf{x}' \mathbf{M}_k \mathbf{x},$$

which is (25). The non-negativity reflects Theorem 6.3. □

Theorem 6.5 ($\bar{R}^2$ Increases iff $|t| > 1$). Let $\mathbf{X}_{k+1} = [\mathbf{X}_k \ \mathbf{x}]$ as in Lemma 6.4, with $\mathbf{x} \notin \mathcal{C}(\mathbf{X}_k)$ (so that $\mathbf{x}'\mathbf{M}_k\mathbf{x} > 0$ and $\mathbf{X}_{k+1}$ has full column rank), and assume $n > k + 2$, $\text{SST} > 0$ and $\text{SSE}_{k+1} > 0$, so that both adjusted R^2 's and the t statistic below are defined. Then adding the regressor $\mathbf{x}$ increases $\bar{R}^2$ if and only if its t -statistic in the enlarged model exceeds 1 in absolute value.

Proof. Let the model with k regressors have $\text{MSE}_k = \text{SSE}_k/(n - k - 1)$, and the model with an additional regressor have $\text{MSE}_{k+1} = \text{SSE}_{k+1}/(n - k - 2)$. Since $\bar{R}^2 = 1 - \text{MSE}/s_Y^2$ with the same $s_Y^2 = \text{SST}/(n - 1) > 0$ in both models, $\bar{R}_{k+1}^2 > \bar{R}_k^2$ if and only if $\text{MSE}_{k+1} < \text{MSE}_k$, i.e.,

$$\frac{\text{SSE}_{k+1}}{n - k - 2} < \frac{\text{SSE}_k}{n - k - 1}. \quad (26)$$

Step 1: Apply Lemma 6.4. By (25), $\text{SSE}_k = \text{SSE}_{k+1} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}$. Substitute this into (26):

$$\begin{aligned} \frac{\text{SSE}_{k+1}}{n - k - 2} &< \frac{\text{SSE}_{k+1} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{n - k - 1} \\ (n - k - 1)\text{SSE}_{k+1} &< (n - k - 2)[\text{SSE}_{k+1} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}] \\ \text{SSE}_{k+1} &< (n - k - 2) \cdot \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}. \end{aligned}$$

Dividing by $\text{SSE}_{k+1} = (n - k - 2)\text{MSE}_{k+1} > 0$:

$$1 < \frac{\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{\text{MSE}_{k+1}}. \quad (27)$$

Step 2: Identify the right-hand side as t^2 . The t -statistic for $\hat{\beta}_{k+1}$ in the enlarged model is $t_{k+1} = \hat{\beta}_{k+1}/\text{SE}(\hat{\beta}_{k+1})$ with $\text{SE}(\hat{\beta}_{k+1})^2 = \text{MSE}_{k+1} \cdot [(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1}$ (defined formally in Theorem 7.1). We claim

$$[(\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}]_{k+1,k+1} = (\mathbf{x}'\mathbf{M}_k\mathbf{x})^{-1}. \quad (28)$$

Indeed, with $\mathbf{G} = (\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}\mathbf{X}'_{k+1}$ we have $\hat{\beta}_{k+1} = \mathbf{G}\mathbf{Y}$ for every $\mathbf{Y}$, so by Lemma 6.4 the last row of $\mathbf{G}$ is $\mathbf{g}' = \mathbf{x}'\mathbf{M}_k/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$; and $\mathbf{G}\mathbf{G}' = (\mathbf{X}'_{k+1}\mathbf{X}_{k+1})^{-1}$, whose last diagonal entry is $\mathbf{g}'\mathbf{g} = \mathbf{x}'\mathbf{M}_k\mathbf{M}_k\mathbf{x}/(\mathbf{x}'\mathbf{M}_k\mathbf{x})^2 = 1/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$. Hence $\text{SE}(\hat{\beta}_{k+1})^2 = \text{MSE}_{k+1}/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$, so

$$t_{k+1}^2 = \frac{\hat{\beta}_{k+1}^2}{\text{SE}(\hat{\beta}_{k+1})^2} = \frac{\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{\text{MSE}_{k+1}}.$$

Substituting into (27) gives $t_{k+1}^2 > 1$, i.e., $|t_{k+1}| > 1$. □

Remark (Logical status of the forward references). The proof refers to one later statement: the form of $\text{SE}(\hat{\beta}_{k+1})$ in Theorem 7.1, which is a *definition* of the t statistic rather than a result whose proof is needed here. It is

used only to *interpret* the ratio $\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}/\text{MSE}_{k+1}$ as t_{k+1}^2 ; equation (27) and the identity (28) are self-contained algebraic consequences of Lemma 6.4. A reader seeking strict linearity can defer Theorem 6.5 until after Section 7.

7 Hypothesis Testing and Confidence Regions

7.1 Individual Coefficient t Test

Theorem 7.1 (Individual t Test). *Under (A1)–(A5), for testing $H_0: \beta_j = \beta_j^0$:*

$$t = \frac{\hat{\beta}_j - \beta_j^0}{\text{SE}(\hat{\beta}_j)} \sim t_{n-k-1} \quad \text{under } H_0, \quad (29)$$

where $\text{SE}(\hat{\beta}_j) = s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$ and $s = \sqrt{\text{MSE}}$.

Proof. Step 1: Marginal distribution of $\hat{\beta}_j$. By Theorem 5.1, $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. The j -th coordinate of a multivariate normal vector is itself normal, with mean β_j and variance equal to the (j, j) -entry of the covariance matrix (this is Theorem 4.1 applied with $\mathbf{A} = \boldsymbol{\iota}'_j$, the j -th standard basis row vector):

$$\hat{\beta}_j = \boldsymbol{\iota}'_j \hat{\boldsymbol{\beta}} \sim N(\beta_j, \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}).$$

Standardizing:

$$Z = \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim N(0, 1).$$

Step 2: Chi-square quantity independent of Z . By Theorem 5.2, $V = \text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ and is independent of $\hat{\boldsymbol{\beta}}$. Since Z is a (deterministic) function of $\hat{\beta}_j$, which is a coordinate of $\hat{\boldsymbol{\beta}}$, and a function of an independent random variable remains independent, Z is independent of V .

Step 3. By definition, if $Z \sim N(0, 1)$ and $V \sim \chi_\nu^2$ are independent, then $Z/\sqrt{V/\nu} \sim t_\nu$. Applying this:

$$\begin{aligned} \frac{Z}{\sqrt{V/(n-k-1)}} &= \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \cdot \frac{1}{\sqrt{\text{SSE}/[\sigma^2(n-k-1)]}} \\ &= \frac{\hat{\beta}_j - \beta_j}{\sqrt{\text{MSE}} \cdot \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{s \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim t_{n-k-1}. \end{aligned}$$

Under H_0 : $\beta_j = \beta_j^0$, replace β_j by β_j^0 . □

7.2 Overall F Test

Lemma 7.2 (Distribution of SSR/σ^2 Under H_0). *Let $\mathbf{J} = \frac{1}{n}\mathbf{1}\mathbf{1}'$ be the projection onto $\mathcal{C}(\mathbf{1})$ of Definition 6.1. Under $H_0: \beta_1 = \dots = \beta_k = 0$ (with (A1)–(A5)):*

$$\frac{\text{SSR}}{\sigma^2} \sim \chi_k^2, \quad \text{and SSR is independent of SSE.}$$

Proof. Step 1: Express SSR/σ^2 as a quadratic form in standard normals. Two matrix observations are useful:

- $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ by definition of the hat matrix.
- $\bar{Y}\mathbf{1} = \mathbf{J}\mathbf{Y}$, because $\mathbf{J}\mathbf{Y} = n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{Y}) = n^{-1}\mathbf{1} \cdot n\bar{Y} = \bar{Y}\mathbf{1}$.

Therefore

$$\hat{\mathbf{Y}} - \bar{Y}\mathbf{1} = \mathbf{H}\mathbf{Y} - \mathbf{J}\mathbf{Y} = (\mathbf{H} - \mathbf{J})\mathbf{Y},$$

and $\text{SSR} = (\hat{\mathbf{Y}} - \bar{Y}\mathbf{1})'(\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) = \mathbf{Y}'(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})\mathbf{Y}$.

Under H_0 , $\boldsymbol{\beta} = (\beta_0, 0, \dots, 0)'$, so $\mathbf{X}\boldsymbol{\beta} = \beta_0\mathbf{1}$ and $\mathbf{Y} = \beta_0\mathbf{1} + \boldsymbol{\varepsilon}$. We claim $(\mathbf{H} - \mathbf{J})\mathbf{Y} = (\mathbf{H} - \mathbf{J})\boldsymbol{\varepsilon}$: indeed, since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ we have $\mathbf{H}\mathbf{1} = \mathbf{1}$ (by Theorem 2.5(ii) applied to the first column of $\mathbf{X}$), and $\mathbf{J}\mathbf{1} = n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{1}$, so

$$(\mathbf{H} - \mathbf{J})(\beta_0\mathbf{1}) = \beta_0(\mathbf{H}\mathbf{1} - \mathbf{J}\mathbf{1}) = \beta_0(\mathbf{1} - \mathbf{1}) = \mathbf{0},$$

and hence $(\mathbf{H} - \mathbf{J})\mathbf{Y} = (\mathbf{H} - \mathbf{J})(\beta_0\mathbf{1} + \boldsymbol{\varepsilon}) = (\mathbf{H} - \mathbf{J})\boldsymbol{\varepsilon}$.

Let $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma$. Under (A5), $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$. Then

$$\frac{\text{SSR}}{\sigma^2} = \frac{\boldsymbol{\varepsilon}'(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})\boldsymbol{\varepsilon}}{\sigma^2} = \mathbf{z}'(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J})\mathbf{z}.$$

This reduces to $\mathbf{z}'(\mathbf{H} - \mathbf{J})\mathbf{z}$ once we establish in Step 2 that $\mathbf{H} - \mathbf{J}$ is symmetric idempotent, which implies $(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J}) = (\mathbf{H} - \mathbf{J})^2 = \mathbf{H} - \mathbf{J}$.

Step 2: $\mathbf{H} - \mathbf{J}$ is symmetric idempotent with rank k . Symmetry is immediate. For idempotency, we first show

$$\mathbf{HJ} = \mathbf{J} = \mathbf{JH}. \quad (30)$$

Using $\mathbf{H}\mathbf{1} = \mathbf{1}$: $\mathbf{HJ} = n^{-1}\mathbf{H}\mathbf{1}\mathbf{1}' = n^{-1}\mathbf{1}\mathbf{1}' = \mathbf{J}$; by symmetry, $\mathbf{JH} = (\mathbf{HJ})' = \mathbf{J}$. Also $\mathbf{J}^2 = n^{-2}\mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \mathbf{J}$. Then

$$(\mathbf{H} - \mathbf{J})^2 = \mathbf{H}^2 - \mathbf{HJ} - \mathbf{JH} + \mathbf{J}^2 = \mathbf{H} - \mathbf{J} - \mathbf{J} + \mathbf{J} = \mathbf{H} - \mathbf{J}.$$

For the rank, Lemma 2.4 gives

$$\text{rank}(\mathbf{H} - \mathbf{J}) = \text{tr}(\mathbf{H}) - \text{tr}(\mathbf{J}) = (k + 1) - 1 = k,$$

with $\text{tr}(\mathbf{J}) = n^{-1}\text{tr}(\mathbf{1}\mathbf{1}') = 1$.

The Step 1 expression simplifies: $(\mathbf{H} - \mathbf{J})'(\mathbf{H} - \mathbf{J}) = \mathbf{H} - \mathbf{J}$, so $\text{SSR}/\sigma^2 = \mathbf{z}'(\mathbf{H} - \mathbf{J})\mathbf{z} \sim \chi_k^2$ by Theorem 4.3.

Step 3: Independence of SSR and SSE. Apply Theorem 4.4 to $\mathbf{A}_1 = \mathbf{H} - \mathbf{J}$ and $\mathbf{A}_2 = \mathbf{M}$. The required $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$ follows from $\mathbf{HM} = \mathbf{0}$ (Theorem 2.5(iii)) and $\mathbf{JM} = \mathbf{J} - \mathbf{JH} = \mathbf{0}$ by (30):

$$(\mathbf{H} - \mathbf{J})\mathbf{M} = \mathbf{0}.$$

Hence $\text{SSR}/\sigma^2 = \mathbf{z}'(\mathbf{H} - \mathbf{J})\mathbf{z}$ and $\text{SSE}/\sigma^2 = \mathbf{z}'\mathbf{M}\mathbf{z}$ are independent, and so are SSR and SSE. $\square$

Theorem 7.3 (Overall F Test). *Under (A1)–(A5), for testing $H_0: \beta_1 = \beta_2 = \cdots = \beta_k = 0$ (the third expression below presupposes $\text{SST} > 0$, so that R^2 is defined; under (A5) this holds with probability one):*

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{\text{SSR}/k}{\text{SSE}/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)} \sim F_{k, n-k-1} \quad \text{under } H_0. \quad (31)$$

Proof. By Lemma 7.2 and Theorem 5.2, under H_0 : $\text{SSR}/\sigma^2 \sim \chi_k^2$, $\text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$, and they are independent. By the definition of the F distribution:

$$F = \frac{(\text{SSR}/\sigma^2)/k}{(\text{SSE}/\sigma^2)/(n-k-1)} = \frac{\text{SSR}/k}{\text{SSE}/(n-k-1)} = \frac{\text{MSR}}{\text{MSE}} \sim F_{k, n-k-1}.$$

The equivalence $F = \frac{R^2/k}{(1-R^2)/(n-k-1)}$ follows from $R^2 = \text{SSR}/\text{SST}$ and $1 - R^2 = \text{SSE}/\text{SST}$. $\square$

7.3 General Linear Hypothesis

Theorem 7.4 (General F Test). *Under (A1)–(A5), consider $H_0: \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ where $\mathbf{R}$ is a $q \times (k+1)$ matrix of rank q and $\mathbf{r} \in \mathbb{R}^q$. The test statistic*

$$F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q \cdot s^2} \sim F_{q, n-k-1} \quad \text{under } H_0. \quad (32)$$

Proof. Step 1. By Theorem 5.1, $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. By Theorem 4.1:

$$\mathbf{R}\hat{\boldsymbol{\beta}} \sim N(\mathbf{R}\boldsymbol{\beta}, \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}').$$

Under H_0 , $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$, so $\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r} \sim N(\mathbf{0}, \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}')$.

Step 2: A quadratic form in $\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}$ is chi-square. Let $\boldsymbol{\Sigma}_R = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$. $\boldsymbol{\Sigma}_R$ is positive definite: for nonzero $\mathbf{v} \in \mathbb{R}^q$,

$$\mathbf{v}'\boldsymbol{\Sigma}_R\mathbf{v} = (\mathbf{R}'\mathbf{v})'(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{R}'\mathbf{v}) > 0,$$

since $\text{rank}(\mathbf{R}) = q$ makes $\mathbf{R}'\mathbf{v} \neq \mathbf{0}$, and $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite. Let $\boldsymbol{\Sigma}_R^{1/2}$ be its positive definite symmetric square root.

Set $\mathbf{w} = \sigma^{-1} \Sigma_R^{-1/2} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})$. Theorem 4.1 gives $\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_q)$, so $\mathbf{w}'\mathbf{w} \sim \chi_q^2$ by Theorem 4.3:

$$W := \mathbf{w}'\mathbf{w} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})' \Sigma_R^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{\sigma^2} \sim \chi_q^2.$$

Step 3. By Theorem 5.2, $V = \text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$, independent of $\hat{\boldsymbol{\beta}}$; since W is a function of $\hat{\boldsymbol{\beta}}$, W is independent of V . Then:

$$F = \frac{W/q}{V/(n-k-1)} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})' \Sigma_R^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) / (q\sigma^2)}{\text{SSE} / [\sigma^2(n-k-1)]} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})' \Sigma_R^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q \cdot s^2} \sim F_{q, n-k-1}. \quad \square$$

Remark (Special cases). The individual t test (Theorem 7.1) is the case $q = 1$, $\mathbf{R} = \boldsymbol{\iota}_j'$ (the transpose of the j -th standard basis vector), $\mathbf{r} = \beta_j^0$, where $t^2 = F \sim F_{1, n-k-1}$. The overall F test (Theorem 7.3) is the case $\mathbf{R} = [\mathbf{0}_{k \times 1} \mid \mathbf{I}_k]$ (of rank $q = k$), $\mathbf{r} = \mathbf{0}_k$: by Corollary 7.5 below, the statistic (32) then equals $\frac{(\text{SSE}_R - \text{SSE}_U)/k}{\text{SSE}_U/(n-k-1)}$, where the reduced (intercept-only) model has fit $\bar{Y}\mathbf{1}$, so $\text{SSE}_R = \text{SST}$ and $\text{SSE}_R - \text{SSE}_U = \text{SST} - \text{SSE} = \text{SSR}$ by Theorem 6.1; the statistic is therefore MSR/MSE , as in (31).

Corollary 7.5 (Nested F Test via Residual Sum of Squares). *Let SSE_U and SSE_R be the residual sums of squares from the full and reduced models under $H_0: \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ (q constraints, $\text{rank}(\mathbf{R}) = q$). Then the F statistic in Theorem 7.4 is equivalently expressed as*

$$F = \frac{(\text{SSE}_R - \text{SSE}_U)/q}{\text{SSE}_U/(n - k - 1)}. \quad (33)$$

(An algebraic identity using only (A4) and $\text{rank}(\mathbf{R}) = q$; (A1)–(A5) enter only through $F \sim F_{q, n-k-1}$ under H_0 .)

Proof. By Theorem 10.1, the constrained estimator satisfies

$$\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}). \quad (34)$$

Step 1: Decompose SSE_R . Since $\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + \mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R) = \mathbf{e} + \mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)$,

$$\begin{aligned} \text{SSE}_R &= \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2 = \|\mathbf{e}\|^2 + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2 + 2\mathbf{e}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R) \\ &= \text{SSE}_U + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2, \end{aligned}$$

since the cross term vanishes: $\mathbf{e}'\mathbf{X} = \mathbf{0}'$ by Theorem 2.6(i).

Step 2: Simplify the squared norm. Let $\mathbf{A} = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})$ denote the trailing factor in (34),

so that $\hat{\beta} - \hat{\beta}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{A}$. Then

$$\begin{aligned}\|\mathbf{X}(\hat{\beta} - \hat{\beta}_R)\|^2 &= (\hat{\beta} - \hat{\beta}_R)'\mathbf{X}'\mathbf{X}(\hat{\beta} - \hat{\beta}_R) \\ &= [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{A}]'\mathbf{X}'\mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{A}] \\ &= \mathbf{A}'\mathbf{R}\underbrace{(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}}_{=(\mathbf{X}'\mathbf{X})^{-1}}\mathbf{R}'\mathbf{A} \\ &= \mathbf{A}'\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{A}.\end{aligned}$$

Expanding $\mathbf{A}$:

$$\|\mathbf{X}(\hat{\beta} - \hat{\beta}_R)\|^2 = (\mathbf{R}\hat{\beta} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\underbrace{\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}}_{=\mathbf{I}_q}(\mathbf{R}\hat{\beta} - \mathbf{r}).$$

The bracketed product collapses to $\mathbf{I}_q$, leaving

$$\text{SSE}_R - \text{SSE}_U = \|\mathbf{X}(\hat{\beta} - \hat{\beta}_R)\|^2 = (\mathbf{R}\hat{\beta} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\beta} - \mathbf{r}). \quad (35)$$

Step 3: Compare with Theorem 7.4. The numerator in Theorem 7.4's F statistic equals the right-hand side of (35) divided by q . Since the denominator $s^2 = \text{SSE}_U/(n - k - 1)$ is the same, substituting yields

$$F = \frac{(\text{SSE}_R - \text{SSE}_U)/q}{\text{SSE}_U/(n - k - 1)}.$$

□

Remark (Logical status of the forward reference). The proof relies on the constrained estimator formula (34) from Theorem 10.1 in Section 10; Theorem 10.1 is purely algebraic and does not depend on anything in this section, so no circularity arises. A reader seeking strict linearity can defer this corollary until after Section 10. Only for homogeneous constraints $\mathbf{r} = \mathbf{0}$ is there a derivation without constrained OLS: $\mathbf{X}\{\boldsymbol{\beta} : \mathbf{R}\boldsymbol{\beta} = \mathbf{0}\}$ is then a subspace of $\mathcal{C}(\mathbf{X})$ of dimension $k + 1 - q$; with $\mathbf{H}_R$ the orthogonal projection onto it, $\mathbf{H} - \mathbf{H}_R$ is symmetric idempotent of rank q , $\mathbf{Y} = \mathbf{H}_R\mathbf{Y} + (\mathbf{H} - \mathbf{H}_R)\mathbf{Y} + \mathbf{M}\mathbf{Y}$ is an orthogonal decomposition, and $\text{SSE}_R - \text{SSE}_U = \mathbf{Y}'(\mathbf{H} - \mathbf{H}_R)\mathbf{Y}$ by Pythagoras, with $\mathbf{H}_R = \mathbf{H} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. For $\mathbf{r} \neq \mathbf{0}$ the feasible set is affine, not a subspace, and this identity is *false*: $\mathbf{Y}'(\mathbf{H} - \mathbf{H}_R)\mathbf{Y}$ equals $(\mathbf{R}\hat{\boldsymbol{\beta}})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}})$, whereas the correct value (35) is centred at $\mathbf{r}$.

Corollary 7.6 ($t^2 = F$ for a Single Linear Constraint). *For testing $H_0: \mathbf{c}'\boldsymbol{\beta} = r_0$ (a single constraint, $q = 1$, $\mathbf{R} = \mathbf{c}'$ with $\mathbf{c} \neq \mathbf{0}$), the F statistic from Theorem 7.4 equals the square of the t statistic:*

$$t = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}}, \quad F = t^2,$$

and $F = t^2 \sim F_{1,n-k-1}$ under H_0 by Theorem 7.4. The identity $F = t^2$ itself is algebraic (it uses only (A4), through $\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c} > 0$); (A1)–(A5) enter only through the distribution.

Proof. With $q = 1$, $\mathbf{R} = \mathbf{c}'$, $\mathbf{r} = r_0$, the general F statistic becomes:

$$F = \frac{(\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)[\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}]^{-1}(\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)}{1 \cdot s^2} = \left(\frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \right)^2 = t^2. \quad \square$$

7.4 Confidence Intervals and Regions

Theorem 7.7 (Confidence Interval for $\mathbf{c}'\boldsymbol{\beta}$). Under (A1)–(A5), for $\mathbf{c} \neq \mathbf{0}$ (so that $\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c} > 0$ by Theorem 2.2, Step 4), a $(1 - \alpha) \times 100\%$ confidence interval for $\mathbf{c}'\boldsymbol{\beta}$ is

$$\mathbf{c}'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2, n-k-1} \cdot s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}, \quad (36)$$

where $t_{\alpha/2, n-k-1}$ is the upper $\alpha/2$ quantile of t_{n-k-1} .

Proof. By Theorem 4.1 applied to $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$ with $\mathbf{A} = \mathbf{c}'$ (a $1 \times (k+1)$ row vector), $\mathbf{c}'\hat{\boldsymbol{\beta}} \sim N(\mathbf{c}'\boldsymbol{\beta}, \sigma^2\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c})$. Standardizing:

$$Z = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \sim N(0, 1).$$

By Theorem 5.2, $V = \text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ and is independent of $\hat{\boldsymbol{\beta}}$; since Z is a function of $\hat{\boldsymbol{\beta}}$ alone, Z is

independent of V . By definition of the t distribution:

$$\frac{Z}{\sqrt{V/(n-k-1)}} = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \cdot \frac{1}{\sqrt{\text{SSE}/[\sigma^2(n-k-1)]}} = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \sim t_{n-k-1},$$

where the last step uses $s^2 = \text{SSE}/(n-k-1)$. The pivot gives

$$\mathbf{P}\left(-t_{\alpha/2} \leq \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \leq t_{\alpha/2}\right) = 1 - \alpha,$$

and rearranging for $\mathbf{c}'\boldsymbol{\beta}$ yields the interval. □

Theorem 7.8 (Joint Confidence Ellipsoid). *Under (A1)–(A5), a $(1 - \alpha) \times 100\%$ joint confidence region for $\boldsymbol{\beta}$ is*

$$\{\boldsymbol{\beta} \in \mathbb{R}^{k+1} : (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \leq (k+1)s^2 F_{\alpha, k+1, n-k-1}\}. \quad (37)$$

Proof. Step 1: A chi-square quadratic form in $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$. By Theorem 5.1, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \sim N(\mathbf{0}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. Let $(\mathbf{X}'\mathbf{X})^{1/2}$ be the positive definite symmetric square root and define $\mathbf{w} = \sigma^{-1}(\mathbf{X}'\mathbf{X})^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$. Theorem 4.1 gives

$\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_{k+1})$, so $\mathbf{w}'\mathbf{w} \sim \chi_{k+1}^2$ by Theorem 4.3:

$$\mathbf{w}'\mathbf{w} = \frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}{\sigma^2} \sim \chi_{k+1}^2.$$

Step 2: Form the F ratio. This is a function of $\hat{\boldsymbol{\beta}}$, hence independent of $\text{SSE}/\sigma^2 \sim \chi_{n-k-1}^2$ (Theorem 5.2). By definition of F ,

$$\frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/[(k+1)\sigma^2]}{\text{SSE}/[\sigma^2(n-k-1)]} = \frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}{(k+1)s^2} \sim F_{k+1, n-k-1}.$$

The confidence region collects the $\boldsymbol{\beta}$ for which this statistic is at most $F_{\alpha, k+1, n-k-1}$. □

8 Partitioned Regression

8.1 The Frisch–Waugh–Lovell Theorem

Theorem 8.1 (Frisch–Waugh–Lovell). *Partition $\mathbf{X} = [\mathbf{X}_1 \quad \mathbf{X}_2]$ and $\boldsymbol{\beta} = (\boldsymbol{\beta}'_1, \boldsymbol{\beta}'_2)'$, so the model is $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$. Let $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}'_2\mathbf{X}_2)^{-1}\mathbf{X}'_2$ be the residual maker for $\mathbf{X}_2$. Then:*

- (i) $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'_1\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{M}_2\mathbf{Y}$.
- (ii) *The OLS coefficient vector from regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$ is the $\hat{\boldsymbol{\beta}}_1$ of (i), and the residual vector of that regression equals the residual vector $\mathbf{e}$ from the full regression.*

Proof. (i). The normal equations for the full model are:

$$\begin{pmatrix} \mathbf{X}'_1 \mathbf{X}_1 & \mathbf{X}'_1 \mathbf{X}_2 \\ \mathbf{X}'_2 \mathbf{X}_1 & \mathbf{X}'_2 \mathbf{X}_2 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{X}'_1 \mathbf{Y} \\ \mathbf{X}'_2 \mathbf{Y} \end{pmatrix}.$$

The second block equation reads $\mathbf{X}'_2 \mathbf{X}_1 \hat{\beta}_1 + \mathbf{X}'_2 \mathbf{X}_2 \hat{\beta}_2 = \mathbf{X}'_2 \mathbf{Y}$, which rearranges to

$$\hat{\beta}_2 = (\mathbf{X}'_2 \mathbf{X}_2)^{-1} \mathbf{X}'_2 (\mathbf{Y} - \mathbf{X}_1 \hat{\beta}_1).$$

Substituting this into the first block equation $\mathbf{X}'_1 \mathbf{X}_1 \hat{\beta}_1 + \mathbf{X}'_1 \mathbf{X}_2 \hat{\beta}_2 = \mathbf{X}'_1 \mathbf{Y}$:

$$\mathbf{X}'_1 \mathbf{X}_1 \hat{\beta}_1 + \underbrace{\mathbf{X}'_1 \mathbf{X}_2 (\mathbf{X}'_2 \mathbf{X}_2)^{-1} \mathbf{X}'_2 (\mathbf{Y} - \mathbf{X}_1 \hat{\beta}_1)}_{= \mathbf{X}'_1 \mathbf{H}_2} = \mathbf{X}'_1 \mathbf{Y}, \quad (38)$$

where we recognize the projection onto $\mathcal{C}(\mathbf{X}_2)$:

$$\mathbf{H}_2 := \mathbf{X}_2 (\mathbf{X}'_2 \mathbf{X}_2)^{-1} \mathbf{X}'_2 = \mathbf{I} - \mathbf{M}_2.$$

Expanding (38) using $\mathbf{X}'_1 \mathbf{H}_2 = \mathbf{X}'_1 - \mathbf{X}'_1 \mathbf{M}_2$:

$$\mathbf{X}'_1 \mathbf{X}_1 \hat{\beta}_1 + \mathbf{X}'_1 (\mathbf{Y} - \mathbf{X}_1 \hat{\beta}_1) - \mathbf{X}'_1 \mathbf{M}_2 (\mathbf{Y} - \mathbf{X}_1 \hat{\beta}_1) = \mathbf{X}'_1 \mathbf{Y}.$$

The first two terms combine to $\mathbf{X}'_1 \mathbf{Y}$, cancelling the right-hand side, leaving $\mathbf{X}'_1 \mathbf{M}_2 (\mathbf{Y} - \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1) = \mathbf{0}$, or

$$\mathbf{X}'_1 \mathbf{M}_2 \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 = \mathbf{X}'_1 \mathbf{M}_2 \mathbf{Y}.$$

Since $\mathbf{X}'_1 \mathbf{M}_2 \mathbf{X}_1 = (\mathbf{M}_2 \mathbf{X}_1)' (\mathbf{M}_2 \mathbf{X}_1)$ is positive definite (columns of $\mathbf{M}_2 \mathbf{X}_1$ are linearly independent by (A4)):

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'_1 \mathbf{M}_2 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{M}_2 \mathbf{Y}.$$

(ii). Let $\tilde{\mathbf{Y}} = \mathbf{M}_2 \mathbf{Y}$ and $\tilde{\mathbf{X}}_1 = \mathbf{M}_2 \mathbf{X}_1$. Using $\mathbf{M}'_2 = \mathbf{M}_2 = \mathbf{M}_2^2$, $(\tilde{\mathbf{X}}_1' \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1' \tilde{\mathbf{Y}} = (\mathbf{X}'_1 \mathbf{M}_2 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{M}_2 \mathbf{Y} = \hat{\boldsymbol{\beta}}_1$ by (i). The residual from regressing $\tilde{\mathbf{Y}}$ on $\tilde{\mathbf{X}}_1$ is

$$\tilde{\mathbf{e}} = \mathbf{M}_2 \mathbf{Y} - \mathbf{M}_2 \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 = \mathbf{M}_2 (\mathbf{Y} - \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1).$$

Substituting $\mathbf{Y} - \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 = \mathbf{X}_2 \hat{\boldsymbol{\beta}}_2 + \mathbf{e}$ from the full regression and using $\mathbf{M}_2 \mathbf{X}_2 = \mathbf{0}$:

$$\tilde{\mathbf{e}} = \mathbf{M}_2 \mathbf{X}_2 \hat{\boldsymbol{\beta}}_2 + \mathbf{M}_2 \mathbf{e} = \mathbf{M}_2 \mathbf{e}. \quad (39)$$

Finally, $\mathbf{M}_2 \mathbf{e} = \mathbf{e}$: since $\mathbf{X}' \mathbf{e} = \mathbf{0}$ (Theorem 2.6(i)) includes $\mathbf{X}'_2 \mathbf{e} = \mathbf{0}$,

$$\mathbf{M}_2 \mathbf{e} = \mathbf{e} - \mathbf{X}_2 (\mathbf{X}'_2 \mathbf{X}_2)^{-1} \mathbf{X}'_2 \mathbf{e} = \mathbf{e}.$$

Hence $\tilde{\mathbf{e}} = \mathbf{e}$. □

Remark (The key identity $\mathbf{M}_2\mathbf{M} = \mathbf{M}$). The computation above rests on the identity $\mathbf{M}_2\mathbf{e} = \mathbf{e}$, which can be written equivalently as $\mathbf{M}_2\mathbf{M} = \mathbf{M}$ (applying both sides to $\mathbf{Y}$). This holds because $\mathcal{C}(\mathbf{X}_2) \subseteq \mathcal{C}(\mathbf{X})$, so the orthogonal complement $\mathcal{C}(\mathbf{X})^\perp \subseteq \mathcal{C}(\mathbf{X}_2)^\perp$: anything already orthogonal to $\mathbf{X}$ is automatically orthogonal to $\mathbf{X}_2$, and $\mathbf{M}_2$ leaves it unchanged. This identity is the abstract reason why “residualizing with respect to a subset of regressors” preserves the full-model residual.

8.2 Omitted Variable Bias

Theorem 8.2 (Omitted Variable Bias). *Suppose the true model is $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ with $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (assumption (A2)), but the researcher estimates the “short” regression $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\gamma} + \mathbf{u}$, obtaining $\tilde{\boldsymbol{\gamma}} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{Y}$. Then:*

$$\mathbb{E}[\tilde{\boldsymbol{\gamma}}] = \boldsymbol{\beta}_1 + \underbrace{(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2}_{\text{auxiliary regression coefficients}} \boldsymbol{\beta}_2. \quad (40)$$

The estimator $\tilde{\boldsymbol{\gamma}}$ is unbiased for $\boldsymbol{\beta}_1$ if and only if $\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$, which holds when either $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$ ($\mathbf{X}_1$ and $\mathbf{X}_2$ are orthogonal) or $\boldsymbol{\beta}_2 = \mathbf{0}$ (the omitted variables are irrelevant).

Proof. Substituting the true DGP $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ into the short-regression estimator:

$$\begin{aligned} \tilde{\boldsymbol{\gamma}} &= (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{Y} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}) \\ &= \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\boldsymbol{\varepsilon}. \end{aligned}$$

Taking expectations (using $E[\epsilon] = \mathbf{0}$):

$$E[\tilde{\gamma}] = \beta_1 + (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \beta_2.$$

□

Remark (Two-factor structure of the bias). The bias term $(\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \beta_2$ is the product of two conceptually distinct quantities:

- $(\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2$ is the matrix of OLS coefficients from regressing each column of $\mathbf{X}_2$ on $\mathbf{X}_1$ —an *auxiliary regression* describing how the omitted variables relate to the included ones.
- β_2 is the effect of the omitted variables on $\mathbf{Y}$ in the true model.

The bias vanishes if either factor is zero: if $\mathbf{X}_1$ and $\mathbf{X}_2$ are orthogonal ($\mathbf{X}'_1 \mathbf{X}_2 = \mathbf{0}$), omission is harmless; if the omitted variables have no effect ($\beta_2 = \mathbf{0}$), there is nothing to omit. Both ingredients must be nonzero to generate bias.

Remark (Connection to FWL). Theorem 8.1 shows that the correct (full-model) coefficient $\hat{\beta}_1$ is obtained by regressing $\mathbf{M}_2 \mathbf{Y}$ on $\mathbf{M}_2 \mathbf{X}_1$ —that is, after partialling $\mathbf{X}_2$ out of both $\mathbf{Y}$ and $\mathbf{X}_1$. The short regression instead regresses $\mathbf{Y}$ directly on $\mathbf{X}_1$, effectively replacing $\mathbf{M}_2$ with $\mathbf{I}$. The two coefficient vectors differ by an exact sample identity: the first block of the full-model normal equations, $\mathbf{X}'_1 \mathbf{X}_1 \hat{\beta}_1 + \mathbf{X}'_1 \mathbf{X}_2 \hat{\beta}_2 = \mathbf{X}'_1 \mathbf{Y} = \mathbf{X}'_1 \mathbf{X}_1 \tilde{\gamma}$, gives

$$\tilde{\gamma} - \hat{\beta}_1 = (\mathbf{X}'_1 \mathbf{X}_1)^{-1} \mathbf{X}'_1 \mathbf{X}_2 \hat{\beta}_2,$$

the bias term of (40) with $\hat{\beta}_2$ in place of β_2 ; taking expectations (Theorem 3.2) recovers (40). OVB measures the cost of not partialling out the omitted variables.

9 Prediction

Let $\mathbf{x}_0 \in \mathbb{R}^{k+1}$ be a new observation's regressor vector. The predicted value is $\hat{Y}_0 = \mathbf{x}_0' \hat{\boldsymbol{\beta}}$.

Theorem 9.1 (Confidence Interval for the Conditional Mean). *Under (A1)–(A5), a $(1 - \alpha) \times 100\%$ confidence interval for $E[Y_0 \mid \mathbf{x}_0] = \mathbf{x}_0' \boldsymbol{\beta}$ is*

$$\mathbf{x}_0' \hat{\boldsymbol{\beta}} \pm t_{\alpha/2, n-k-1} \cdot s \sqrt{\mathbf{x}_0' (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0}. \quad (41)$$

Proof. This is the special case $\mathbf{c} = \mathbf{x}_0$ of Theorem 7.7; its hypothesis $\mathbf{c} \neq \mathbf{0}$ holds because the first component of $\mathbf{x}_0$ is the intercept's 1 (Definition 1.1). $\square$

Theorem 9.2 (Prediction Interval for a New Observation). *Under (A1)–(A5), let $Y_0 = \mathbf{x}_0' \boldsymbol{\beta} + \varepsilon_0$ be a new observation whose error $\varepsilon_0 \sim N(0, \sigma^2)$ is independent of $\boldsymbol{\varepsilon}$ (that is, (A3) and (A5) extend to the new observation). Then a $(1 - \alpha) \times 100\%$ prediction interval for Y_0 is*

$$\mathbf{x}_0' \hat{\boldsymbol{\beta}} \pm t_{\alpha/2, n-k-1} \cdot s \sqrt{1 + \mathbf{x}_0' (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0}. \quad (42)$$

Proof. Step 1: Set up. The prediction error is

$$\hat{Y}_0 - Y_0 = \mathbf{x}_0' (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - \varepsilon_0.$$

Here $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \boldsymbol{\varepsilon}$ depends on the training-sample errors $\boldsymbol{\varepsilon}$, whereas ε_0 is the new-observation error,

which by hypothesis is independent of $\boldsymbol{\varepsilon}$; hence $\varepsilon_0 \perp\!\!\!\perp \mathbf{x}'_0(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$. Moreover $(\boldsymbol{\varepsilon}', \varepsilon_0)' \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_{n+1})$ (its MGF factorizes by independence), so

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \\ \mathbf{e} \\ \varepsilon_0 \end{pmatrix} = \begin{pmatrix} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' & \mathbf{0} \\ \mathbf{M} & \mathbf{0} \\ \mathbf{0}' & 1 \end{pmatrix} \begin{pmatrix} \boldsymbol{\varepsilon} \\ \varepsilon_0 \end{pmatrix} \quad (43)$$

is jointly normal (Theorem 4.1), with $\text{cov}(\mathbf{e}, \hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \sigma^2 \mathbf{M}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{0}$ and $\text{cov}(\mathbf{e}, \varepsilon_0) = \mathbf{0}$ (Lemma 3.1(ii)).

Step 2: Compute the variance. Using Step 1's independence and Lemma 3.1 (with $\mathbf{A} = \mathbf{x}'_0$):

$$\begin{aligned} \text{var}(\hat{Y}_0 - Y_0) &= \text{var}(\mathbf{x}'_0(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})) + \text{var}(\varepsilon_0) \\ &= \mathbf{x}'_0 \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{x}_0 + \sigma^2 \\ &= \sigma^2 \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0 + \sigma^2 \\ &= \sigma^2 (1 + \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0). \end{aligned}$$

Step 3: Derive the t pivotal quantity. As a linear function of the jointly normal vector $(\boldsymbol{\varepsilon}', \varepsilon_0)'$, the prediction error is normal (Theorem 4.1):

$$\hat{Y}_0 - Y_0 \sim N(0, \sigma^2(1 + \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0)).$$

For independence from SSE, “ $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \text{SSE}$ and $\varepsilon_0 \perp\!\!\!\perp \text{SSE}$ ” would *not* suffice; what is needed is $\mathbf{e} \perp\!\!\!\perp (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}, \varepsilon_0)$ jointly, which follows from (43) by Theorem 4.2, since both cross-covariances of $\mathbf{e}$ vanish. Hence $\text{SSE} = \mathbf{e}'\mathbf{e}$ is

independent of $\hat{Y}_0 - Y_0$. Replacing σ with s gives a t_{n-k-1} pivot:

$$\frac{\hat{Y}_0 - Y_0}{s\sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}} \sim t_{n-k-1},$$

and inverting yields the prediction interval. □

Remark. The prediction interval is always wider than the confidence interval for the mean, because the prediction variance includes the additional σ^2 term from the irreducible noise ε_0 .

10 Constrained Least Squares and Modeling Remarks

10.1 Remarks on Model Specification

Remark (No-intercept model). When the model omits the intercept, $\mathbf{X}$ has no column of ones. The OLS formula $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ still applies, but Theorem 2.6(ii) is no longer available, so $\sum e_i \neq 0$ in general and the decomposition $\text{SST} = \text{SSR} + \text{SSE}$ may fail. Consequently the two expressions in Definition 6.2 no longer agree and Proposition 6.2 does not apply: $1 - \text{SSE}/\text{SST}$ can be negative, while SSR/SST can exceed 1. (E.g. $n = 2$, $\mathbf{X} = (1, 0)'$, $\mathbf{Y} = (1, 3)'$: $\text{SST} = 2$, $\text{SSR} = 5$, $\text{SSE} = 9$.) Use the uncentered $R^2 = 1 - \text{SSE}/\mathbf{Y}'\mathbf{Y}$ instead (for $\mathbf{Y} \neq \mathbf{0}$), which lies in $[0, 1]$ since $\mathbf{Y}'\mathbf{Y} - \mathbf{Y}'\mathbf{M}\mathbf{Y} = \|\mathbf{H}\mathbf{Y}\|^2 \geq 0$.

Remark (Dummy variable trap). A categorical variable with g levels requires $g - 1$ dummy regressors when the model includes an intercept. Including all g dummies creates the linear dependence $\mathbf{1} = D_1 + \cdots + D_g$, violating (A4).

10.2 Constrained Least Squares

Theorem 10.1 (Constrained OLS). *Consider $\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ subject to $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ (q linear constraints, $\text{rank}(\mathbf{R}) = q$). The constrained estimator is:*

$$\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}), \quad (44)$$

and $\text{SSE}_R \geq \text{SSE}_U$ (where SSE_U is the unconstrained residual sum of squares).

Proof. The objective $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ is a convex quadratic function of $\boldsymbol{\beta}$ (its Hessian $2\mathbf{X}'\mathbf{X}$ is positive definite by (A4)), and the constraint $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ is affine. Hence the KKT conditions are necessary and sufficient for a global minimum.

Step 1: Lagrangian and first-order conditions. The Lagrangian is $\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\lambda}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) + 2\boldsymbol{\lambda}'(\mathbf{R}\boldsymbol{\beta} - \mathbf{r})$ (the factor 2 simplifies later algebra). By Lemma 2.1, differentiating $(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) =$

$\mathbf{Y}'\mathbf{Y} - 2\mathbf{Y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}$ with respect to $\boldsymbol{\beta}'$ gives $-2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}$. The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\beta}'} = -2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X} + 2\boldsymbol{\lambda}'\mathbf{R} = \mathbf{0}', \quad (\text{i})$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\lambda}'} = 2(\mathbf{R}\boldsymbol{\beta} - \mathbf{r})' = \mathbf{0}'. \quad (\text{ii})$$

Transposing (i) gives $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}_R + \mathbf{R}'\hat{\boldsymbol{\lambda}} = \mathbf{X}'\mathbf{Y}$, so

$$\hat{\boldsymbol{\beta}}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}} = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}}. \quad (45)$$

Step 2: Solve for $\hat{\boldsymbol{\lambda}}$. Left-multiply (45) by $\mathbf{R}$ and impose the constraint (ii), $\mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{r}$:

$$\mathbf{r} = \mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}}.$$

Since $\text{rank}(\mathbf{R}) = q$ and $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite, $\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$ is positive definite (hence invertible); see Step 2 of the proof of Theorem 7.4 for the detailed argument. Rearranging:

$$\hat{\boldsymbol{\lambda}} = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}).$$

Substituting back into (45) gives (44).

Step 3: $\text{SSE}_R \geq \text{SSE}_U$. The constrained problem minimizes $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ over $\{\boldsymbol{\beta} : \mathbf{R}\boldsymbol{\beta} = \mathbf{r}\} \subseteq \mathbb{R}^{k+1}$. Since $\hat{\boldsymbol{\beta}}$ minimizes the same objective over the full space, the minimum over a subset cannot be smaller: $\text{SSE}_R \geq \text{SSE}_U$. Equality holds iff $\hat{\boldsymbol{\beta}}$ already satisfies the constraint, i.e., $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{r}$. $\square$